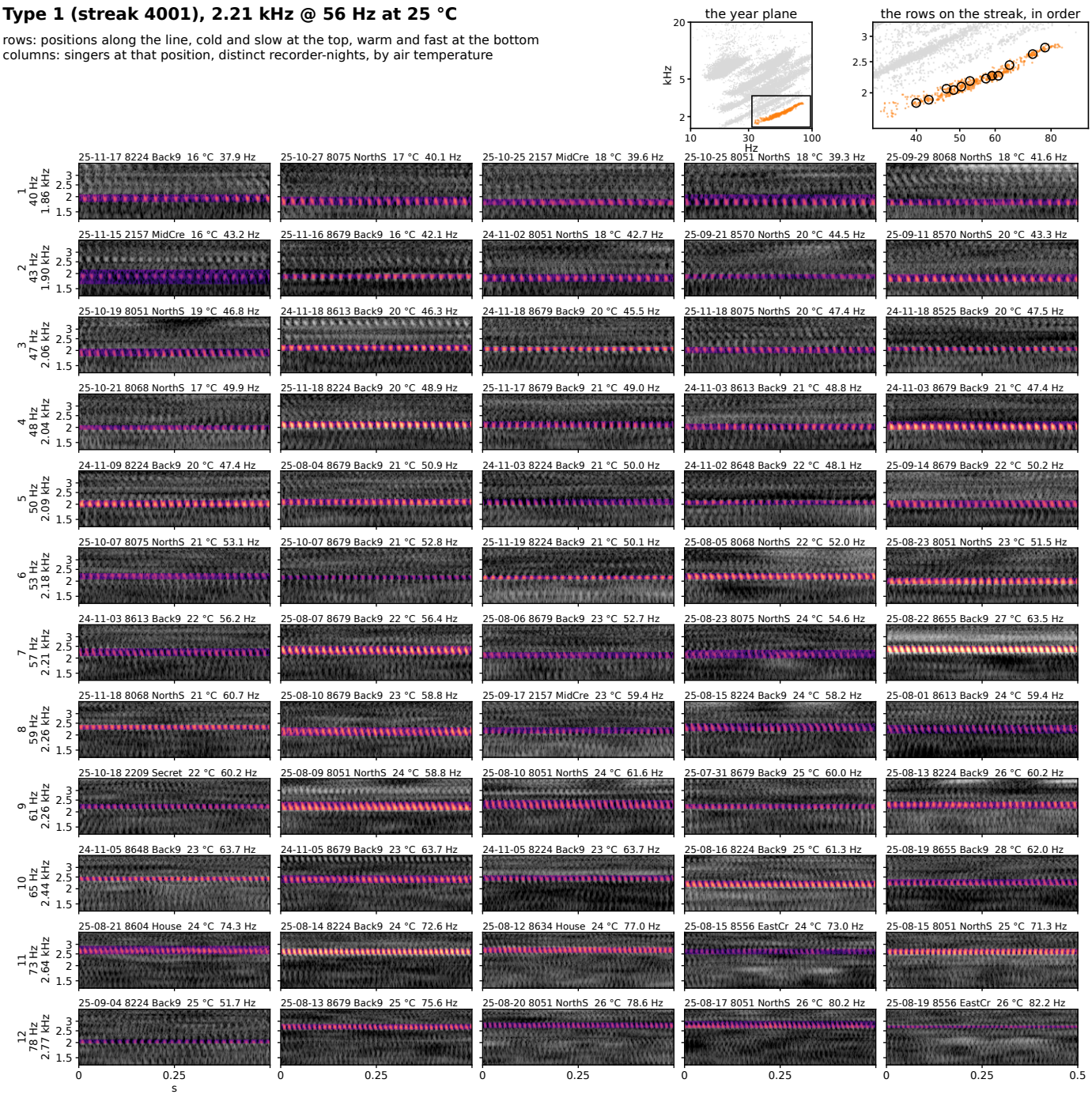

Figure S7-1 (rate-filtered). Type 1 (streak 4001): 2.21 kHz @ 56 Hz at 25 °C, 132 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

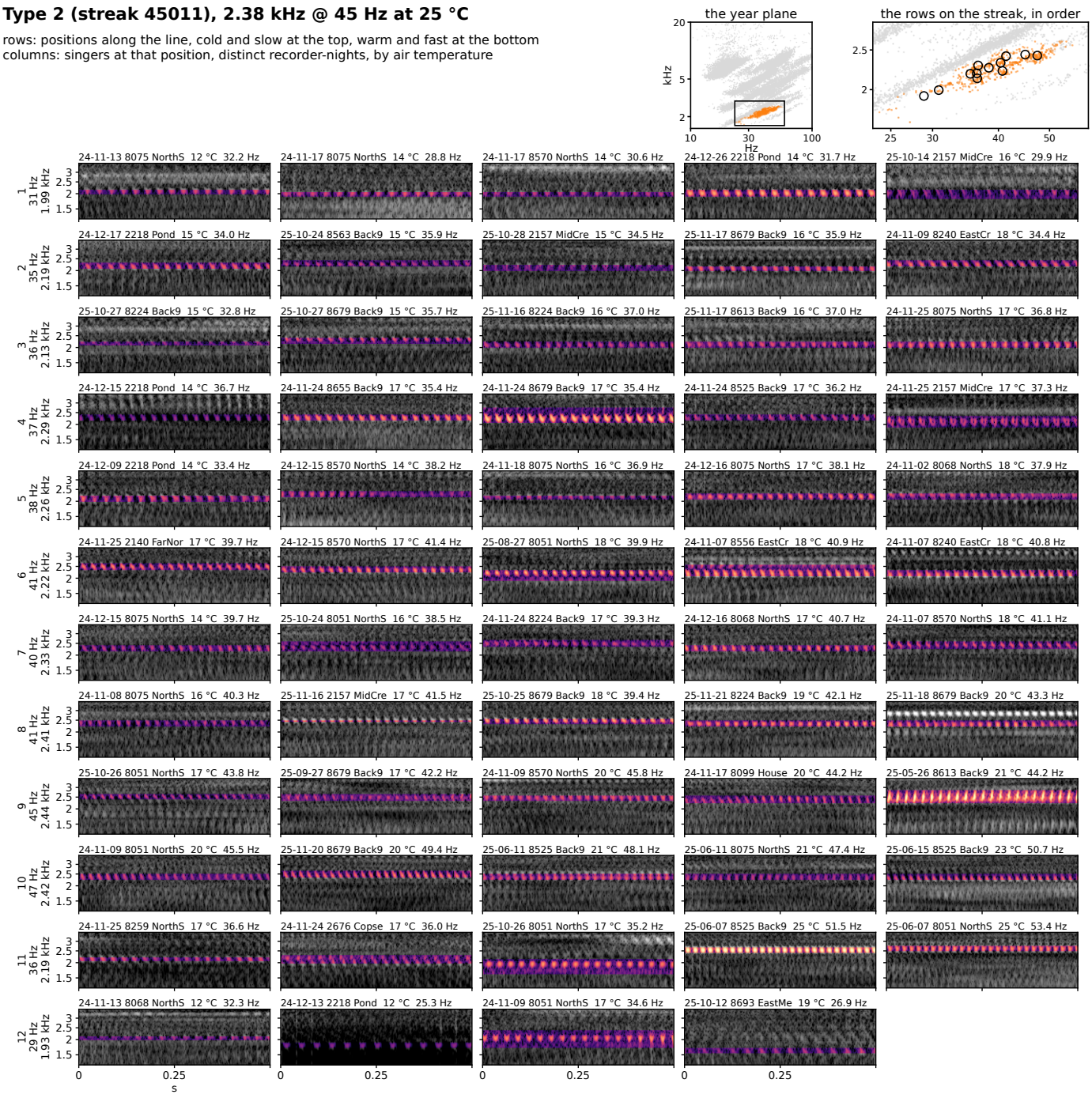

Figure S7-2 (rate-filtered). Type 2 (streak 45011): 2.38 kHz @ 45 Hz at 25 °C, 109 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

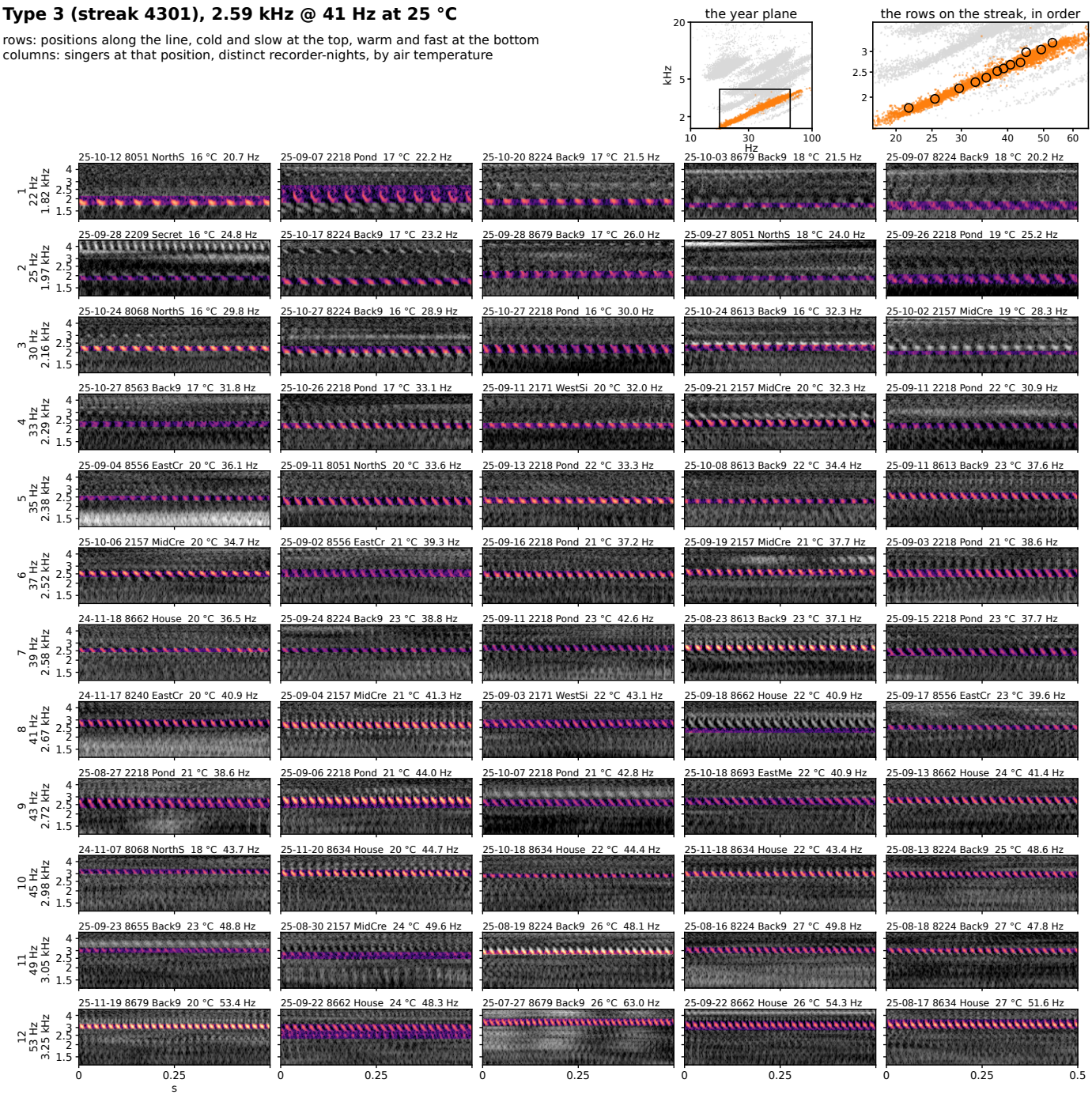

Figure S7-3 (rate-filtered). Type 3 (streak 4301): 2.59 kHz @ 41 Hz at 25 °C, 1655 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

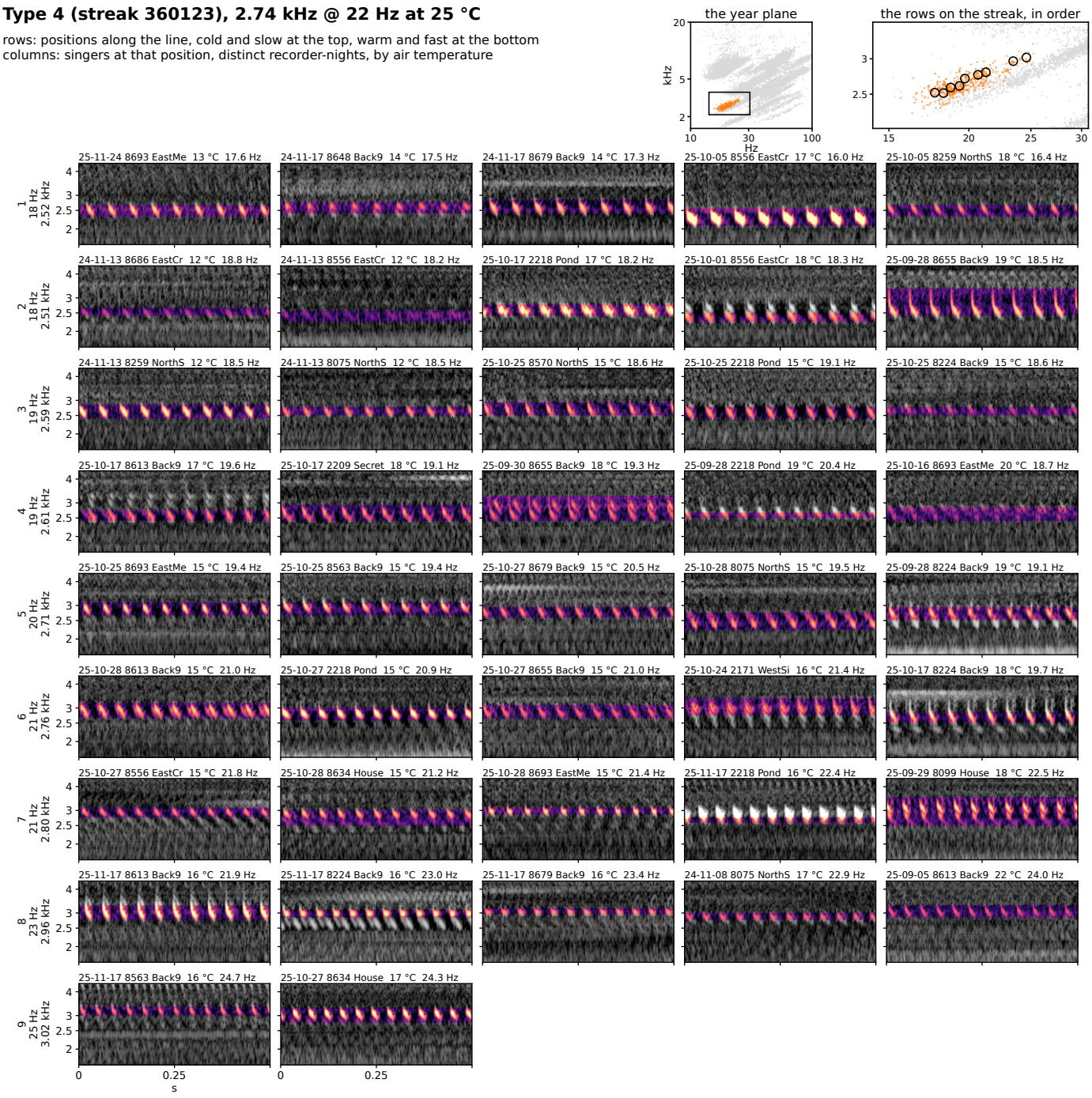

Figure S7-4 (rate-filtered). Type 4 (streak 360123): 2.74 kHz @ 22 Hz at 25 °C, 76 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

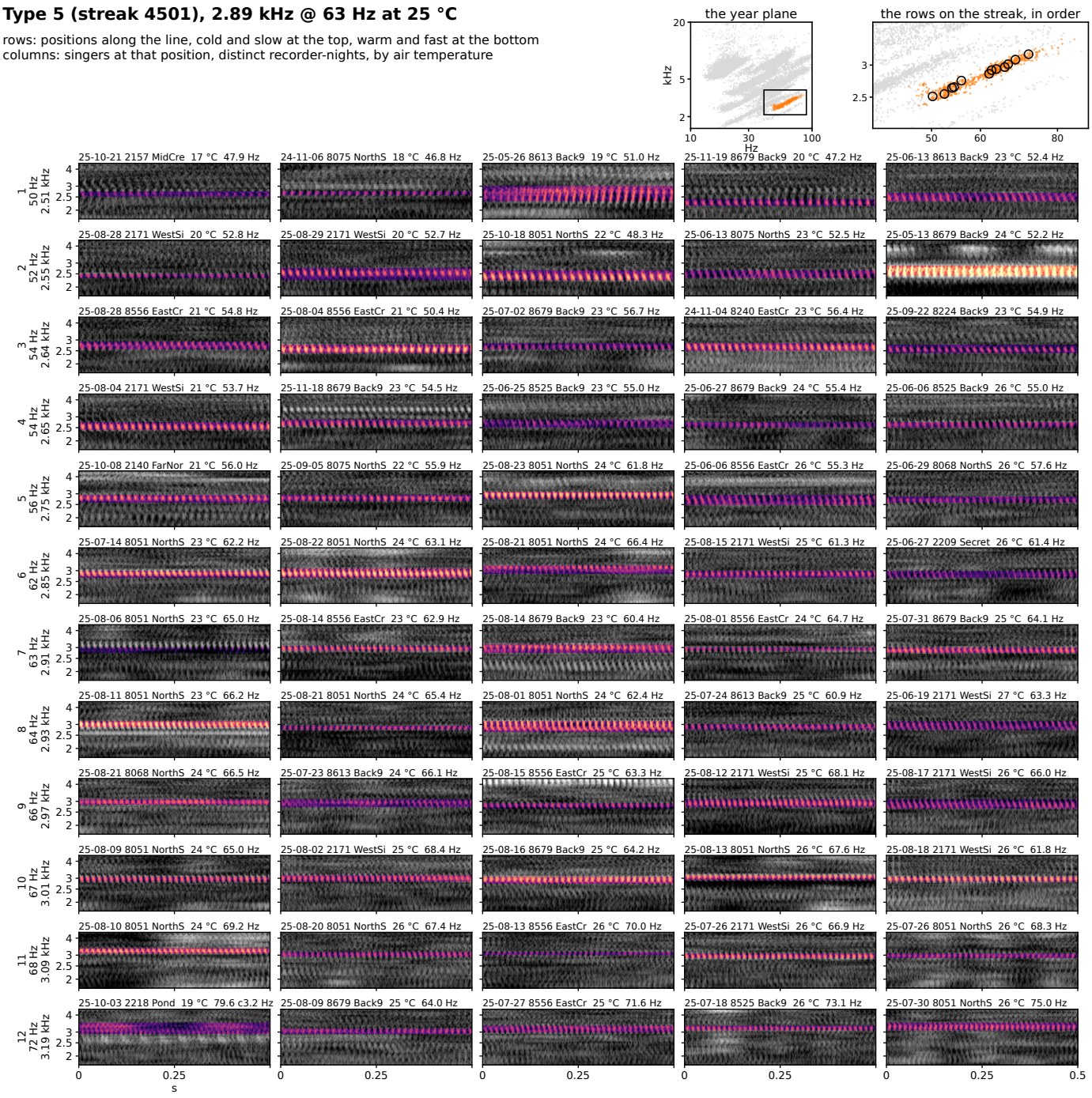

Figure S7-5 (rate-filtered). Type 5 (streak 4501): 2.89 kHz @ 63 Hz at 25 °C, 144 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

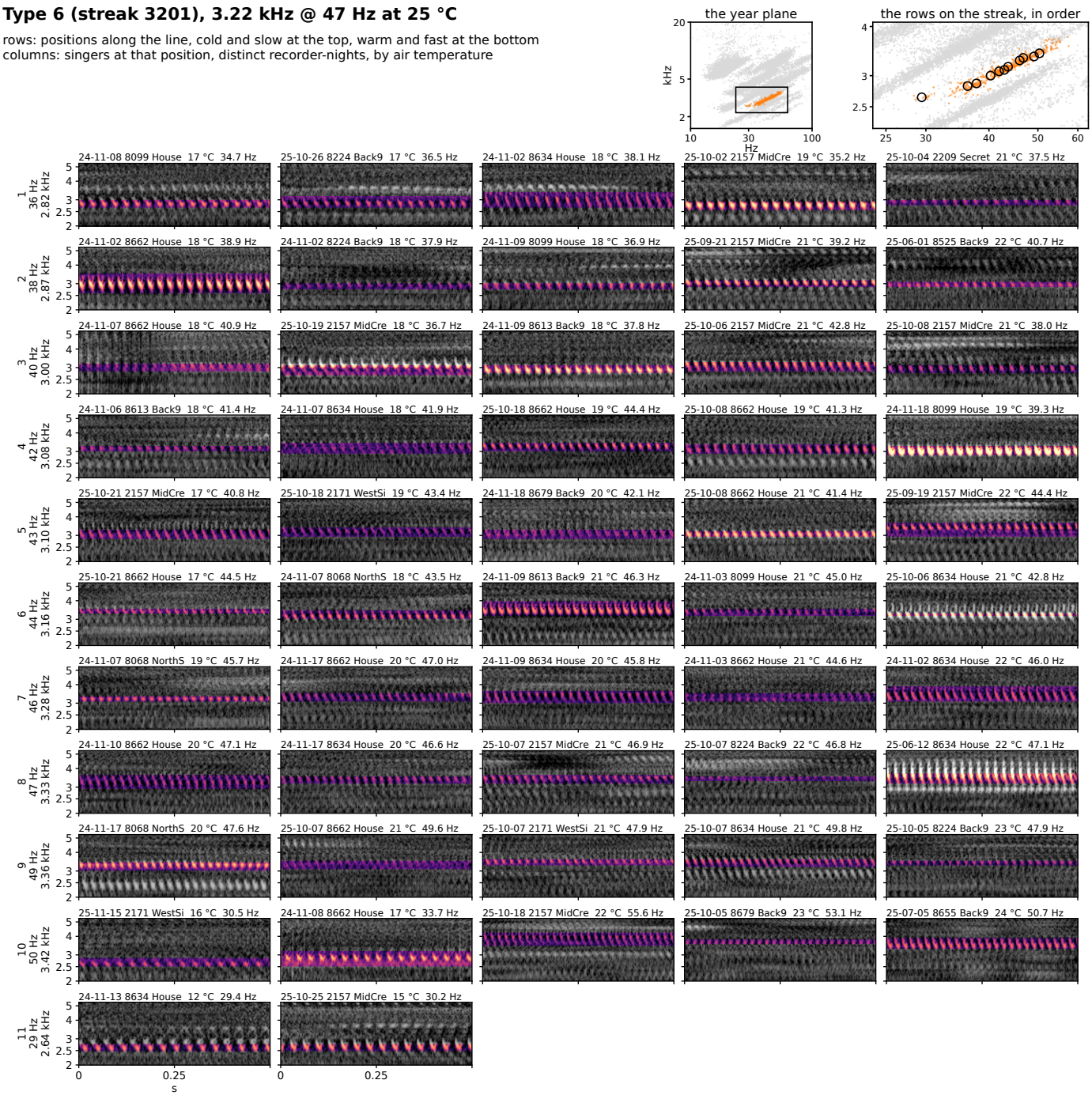

Figure S7-6 (rate-filtered). Type 6 (streak 3201): 3.22 kHz @ 47 Hz at 25 °C, 106 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

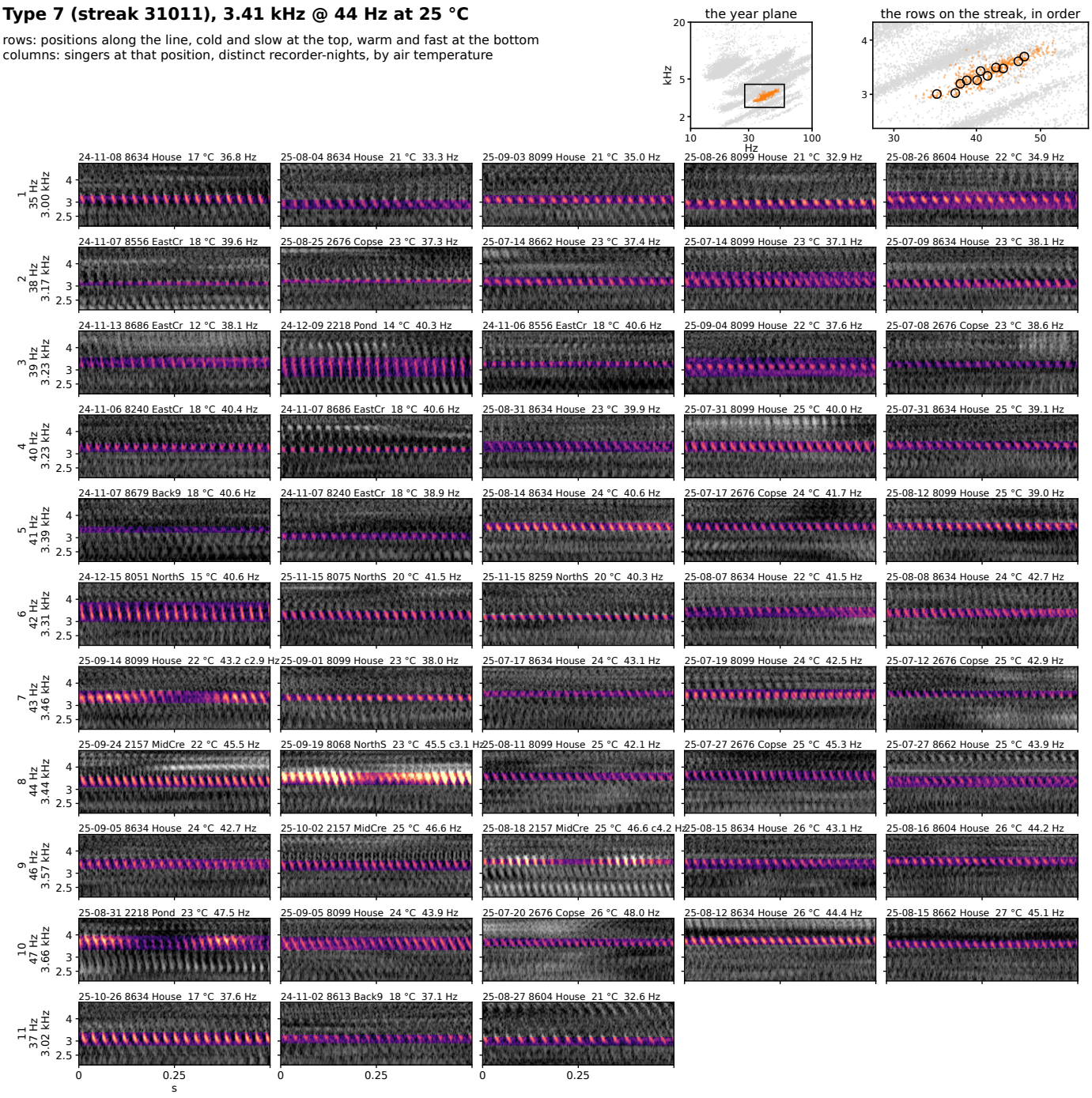

Figure S7-7 (rate-filtered). Type 7 (streak 31011): 3.41 kHz @ 44 Hz at 25 °C, 90 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

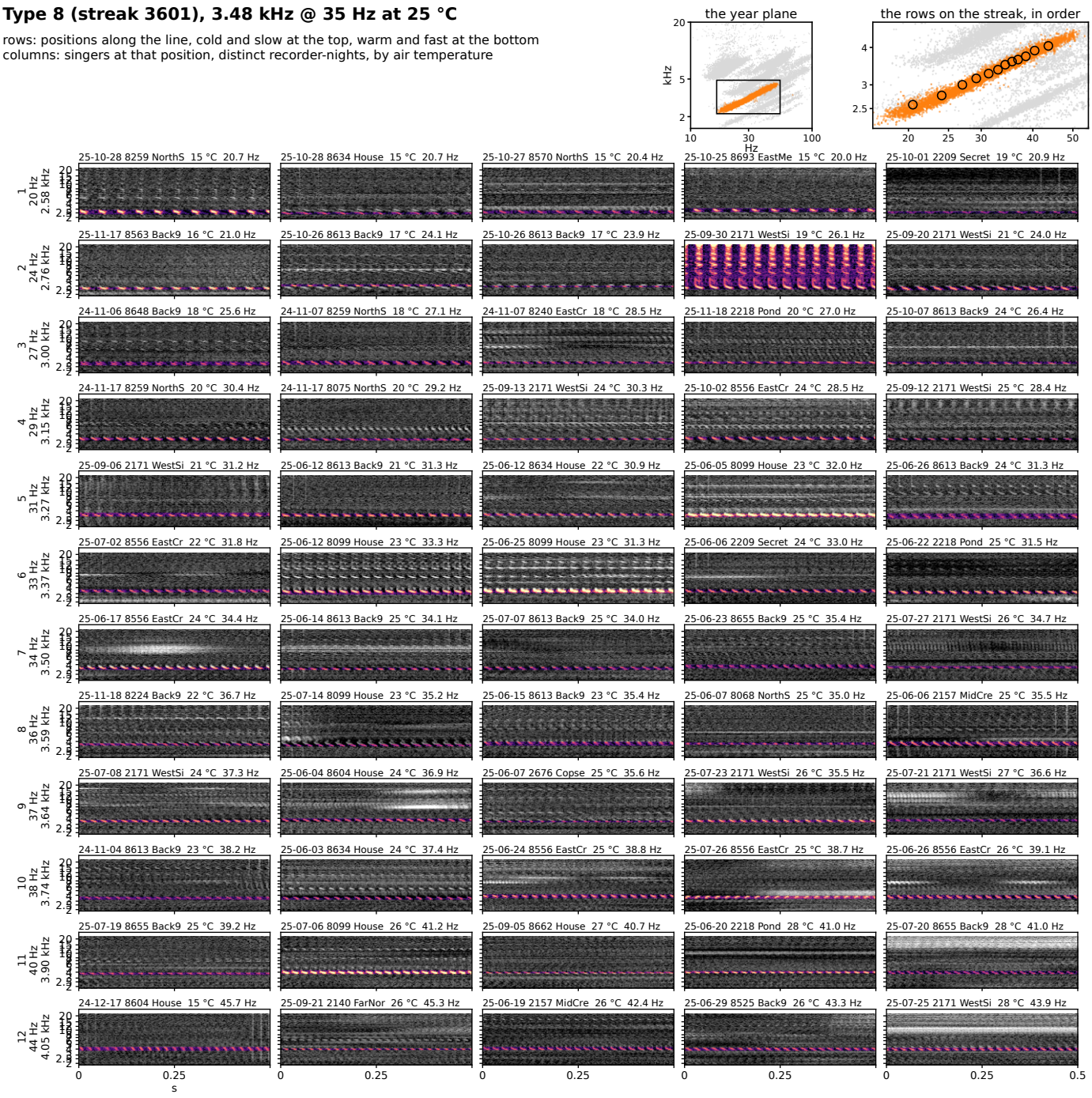

Figure S7-8 (rate-filtered). Type 8 (streak 3601): 3.48 kHz @ 35 Hz at 25 °C, 1834 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

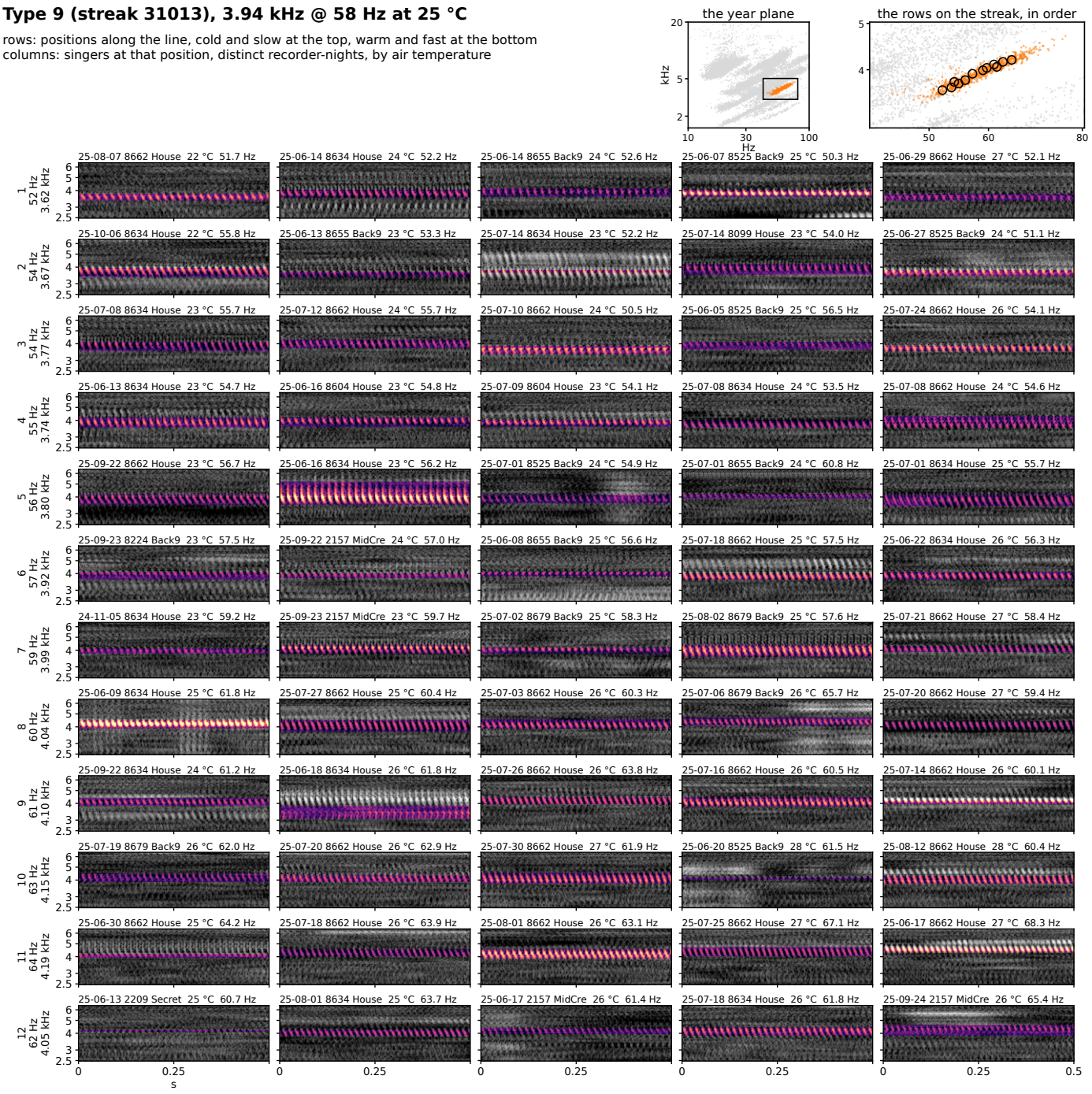

Figure S7-9 (rate-filtered). Type 9 (streak 31013): 3.94 kHz @ 58 Hz at 25 °C, 167 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

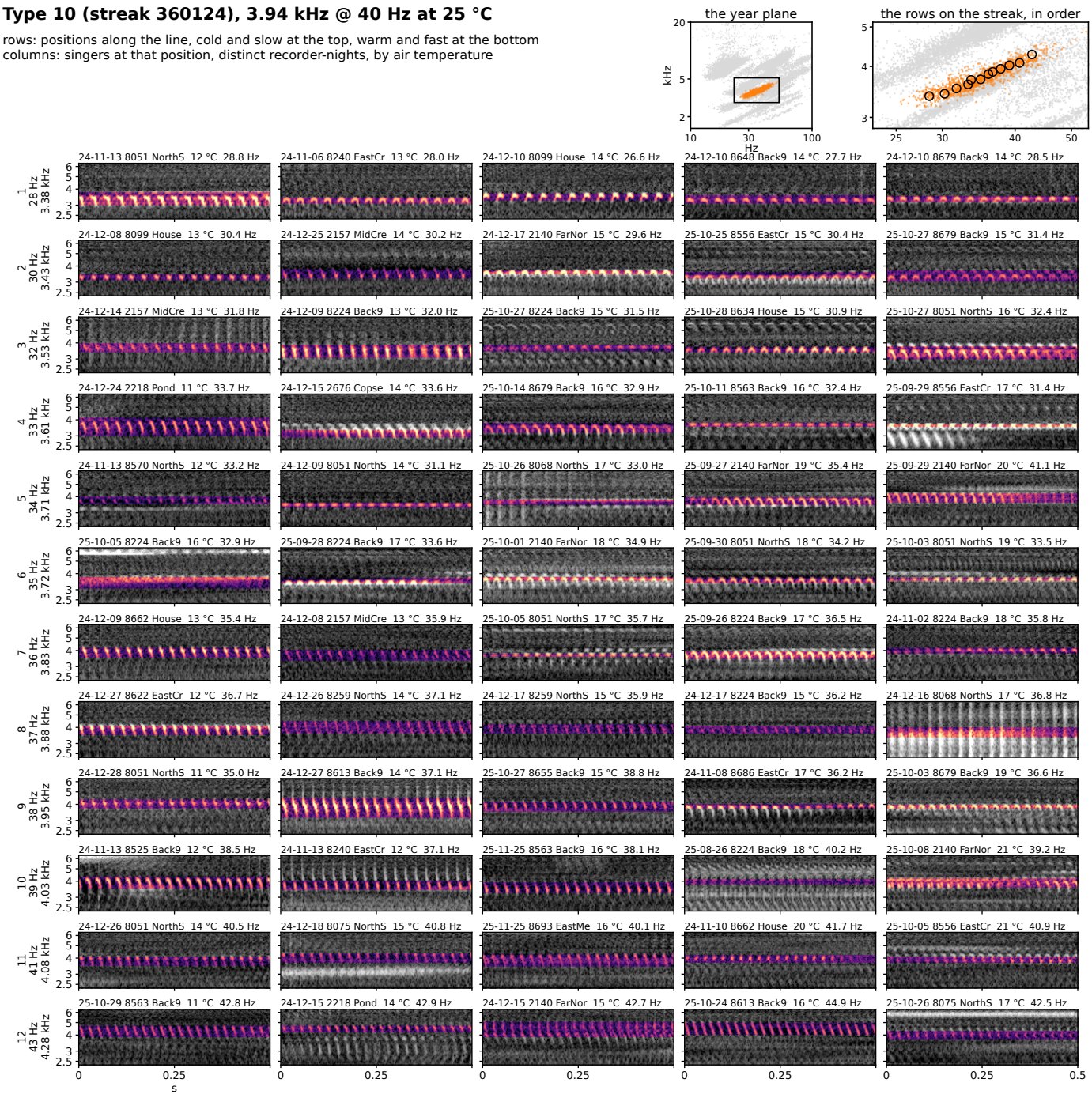

Figure S7-10 (rate-filtered). Type 10 (streak 360124): 3.94 kHz @ 40 Hz at 25 °C, 231 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

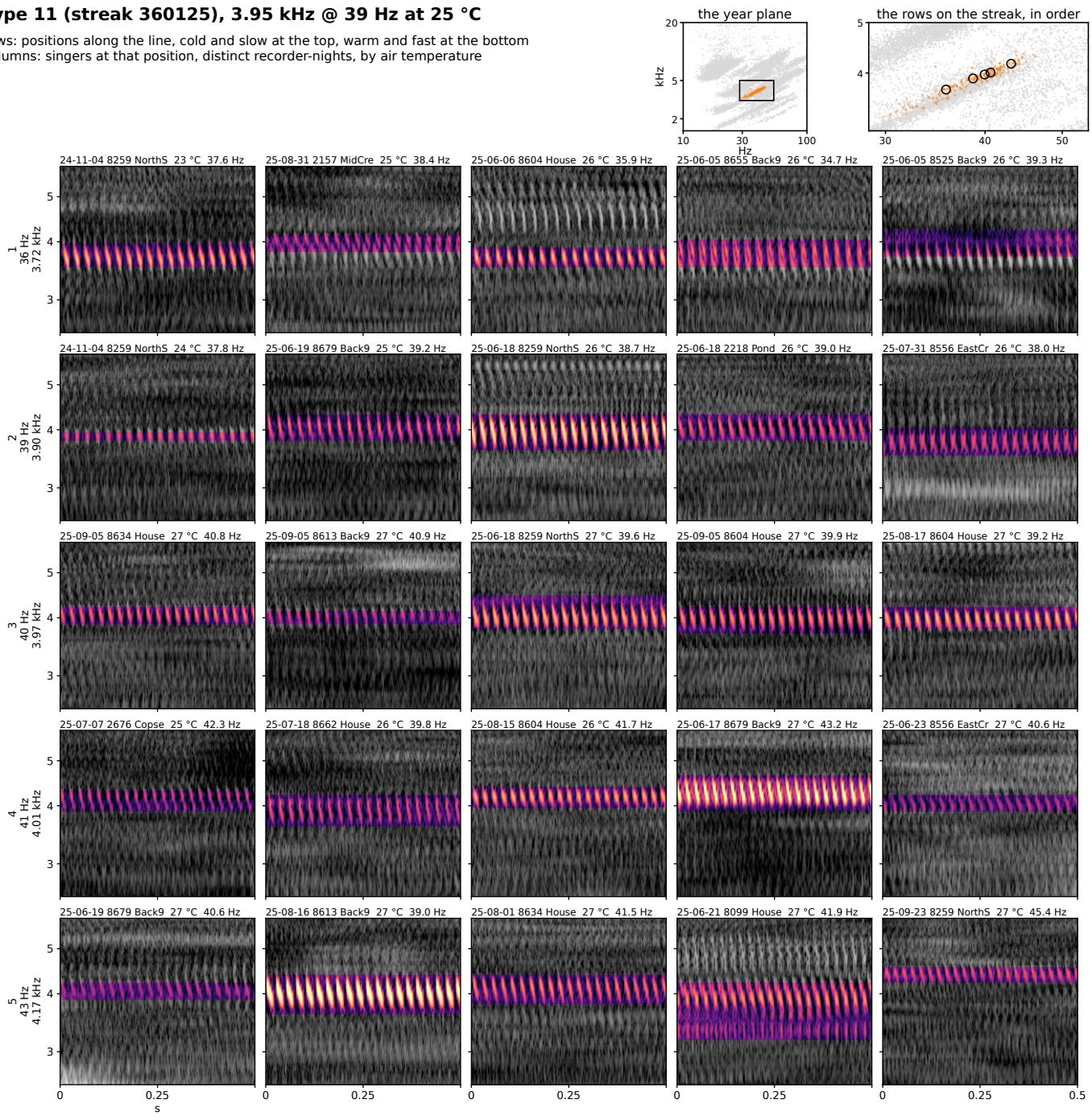

Figure S7-11 (rate-filtered). Type 11 (streak 360125): 3.95 kHz @ 39 Hz at 25 °C, 42 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

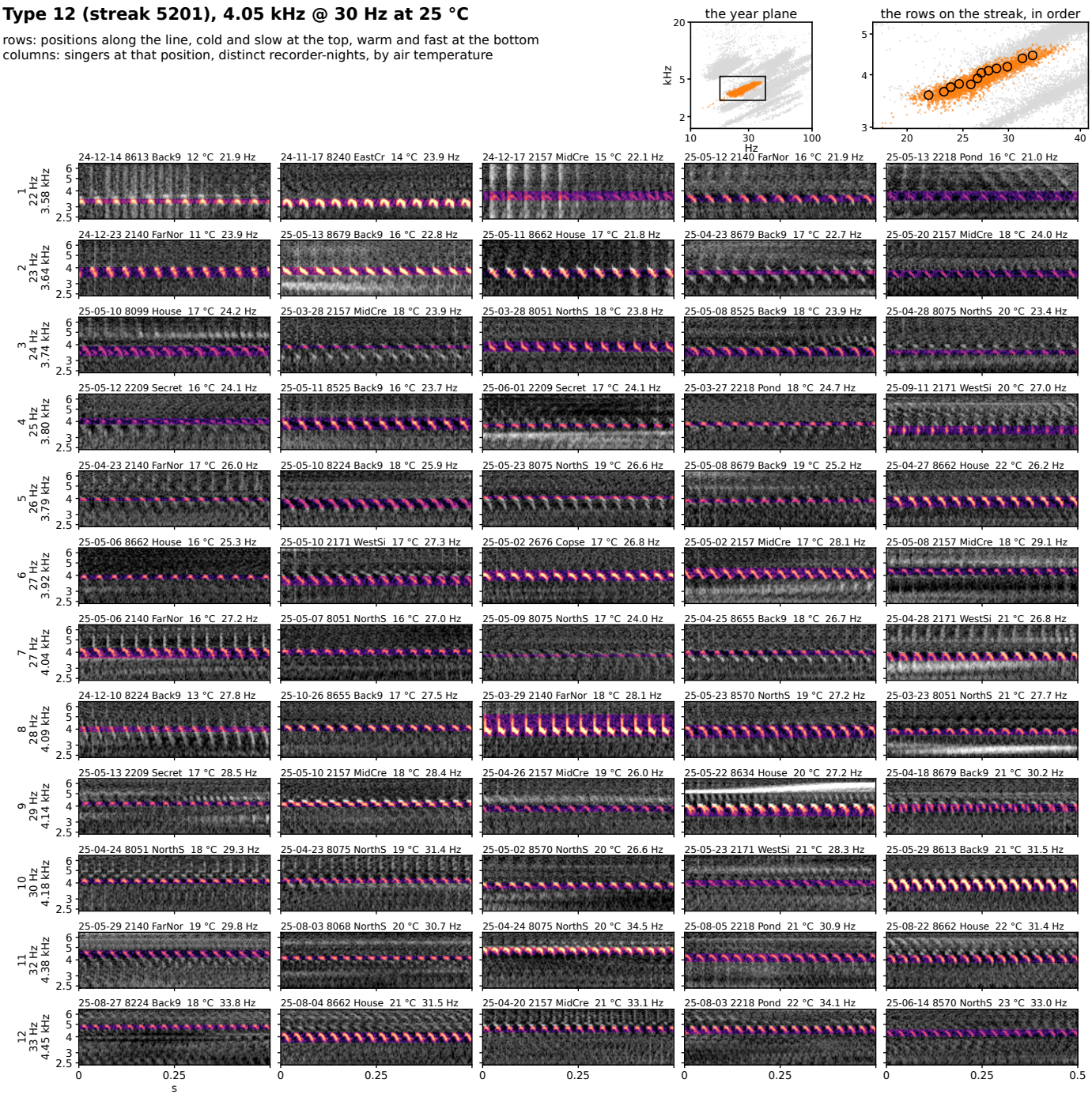

Figure S7-12 (rate-filtered). Type 12 (streak 5201): 4.05 kHz @ 30 Hz at 25 °C, 630 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

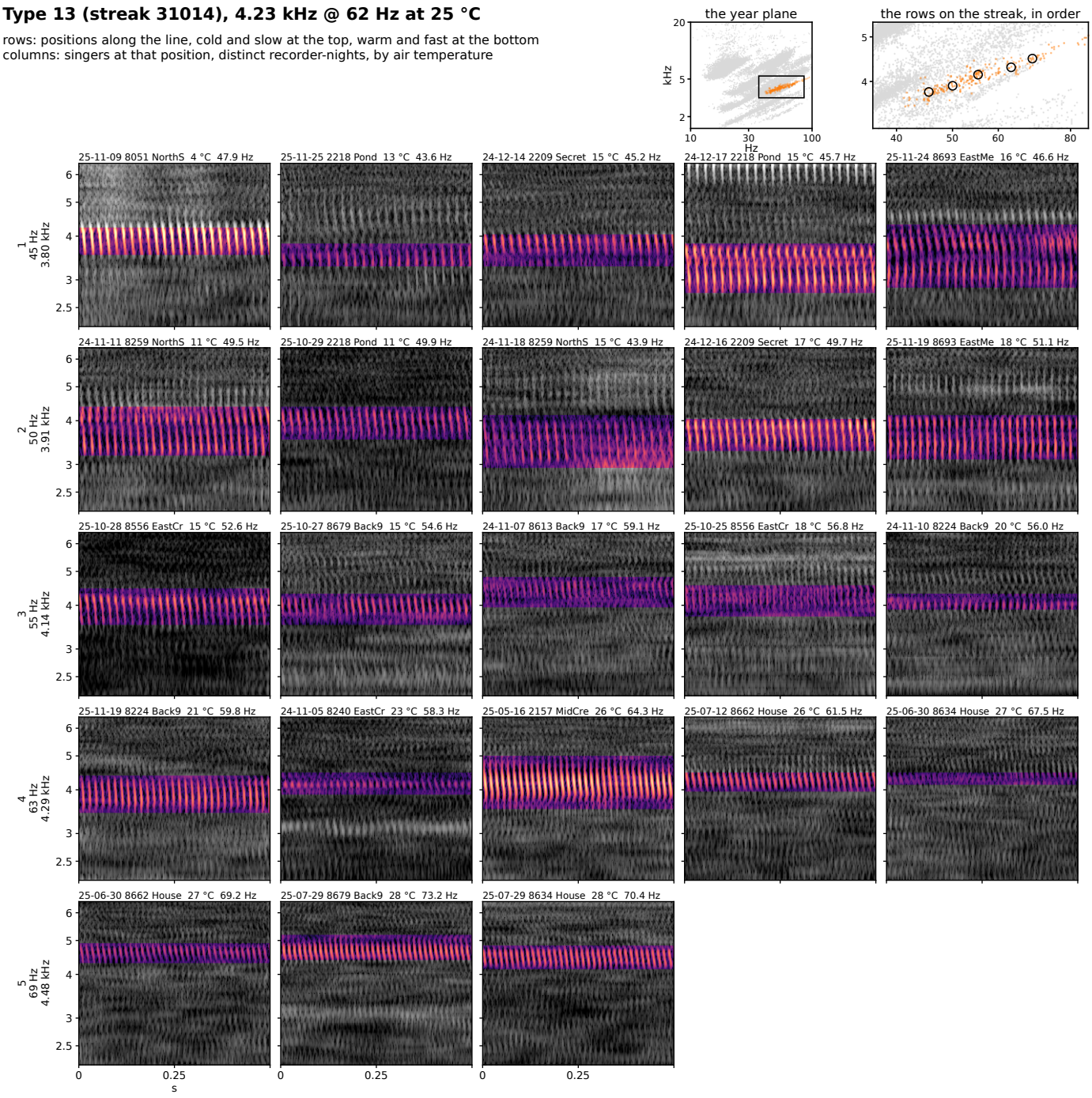

Figure S7-13 (rate-filtered). Type 13 (streak 31014): 4.23 kHz @ 62 Hz at 25 °C, 44 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

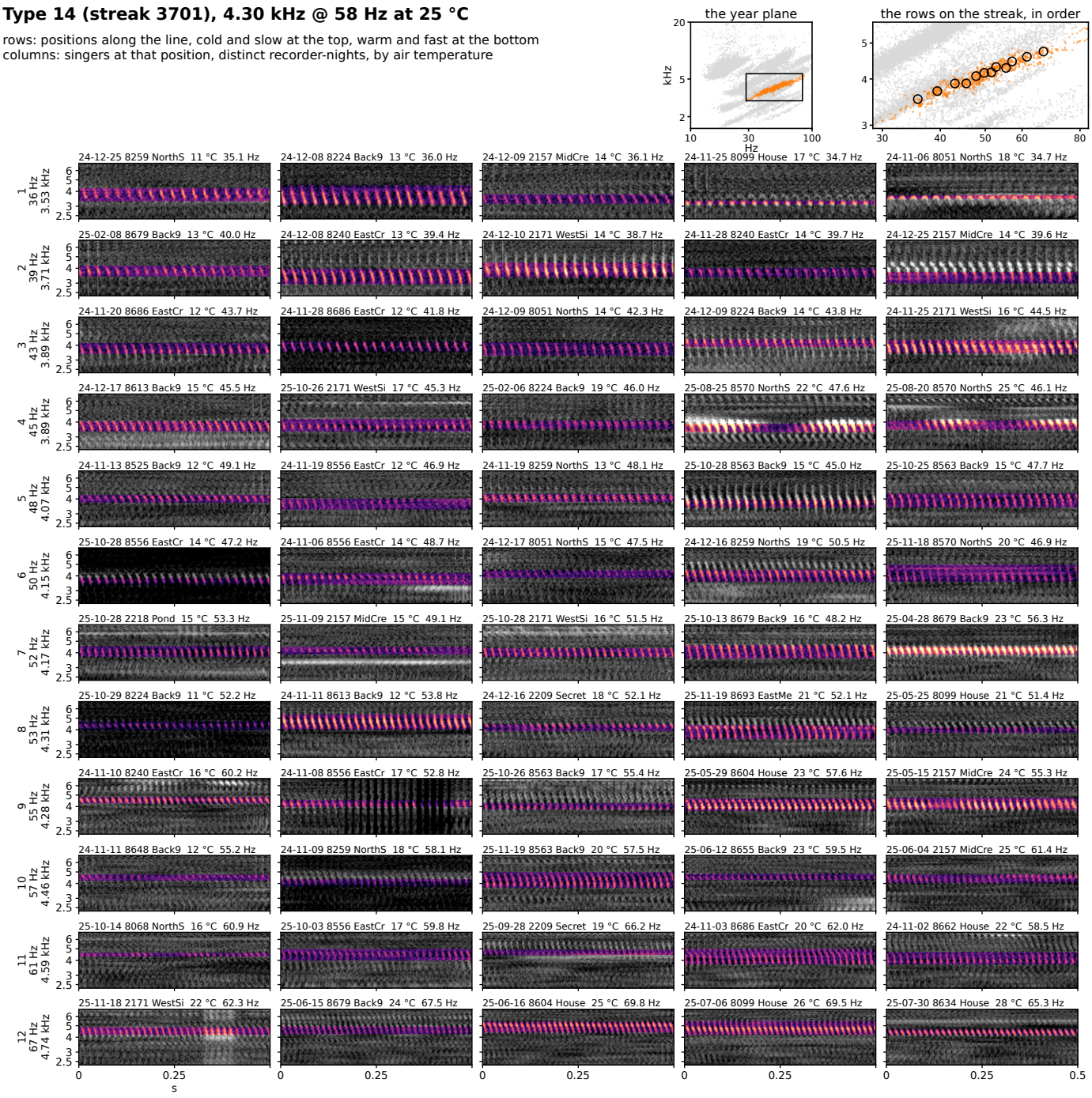

Figure S7-14 (rate-filtered). Type 14 (streak 3701): 4.30 kHz @ 58 Hz at 25 °C, 166 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

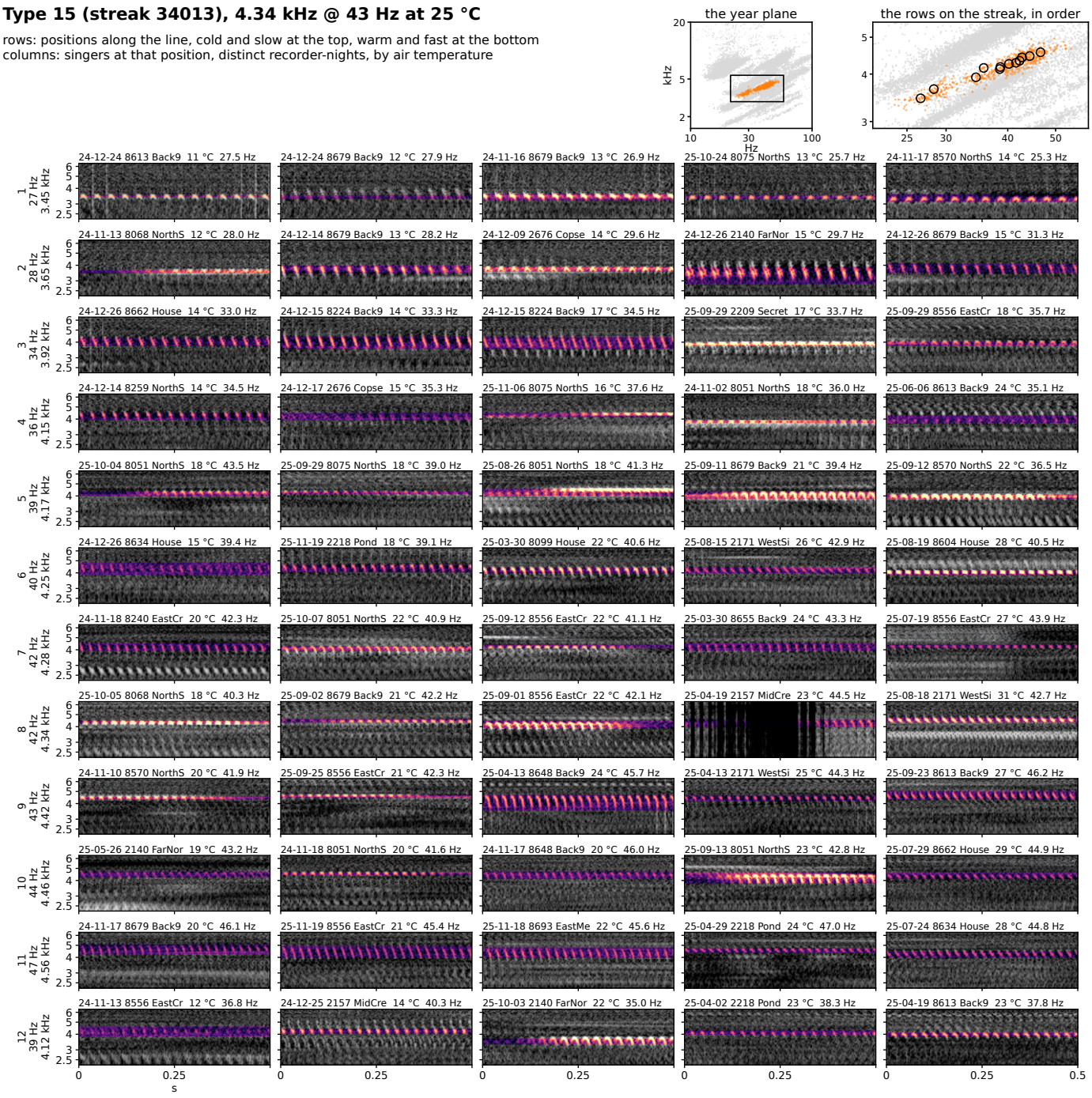

Figure S7-15 (rate-filtered). Type 15 (streak 34013): 4.34 kHz @ 43 Hz at 25 °C, 124 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

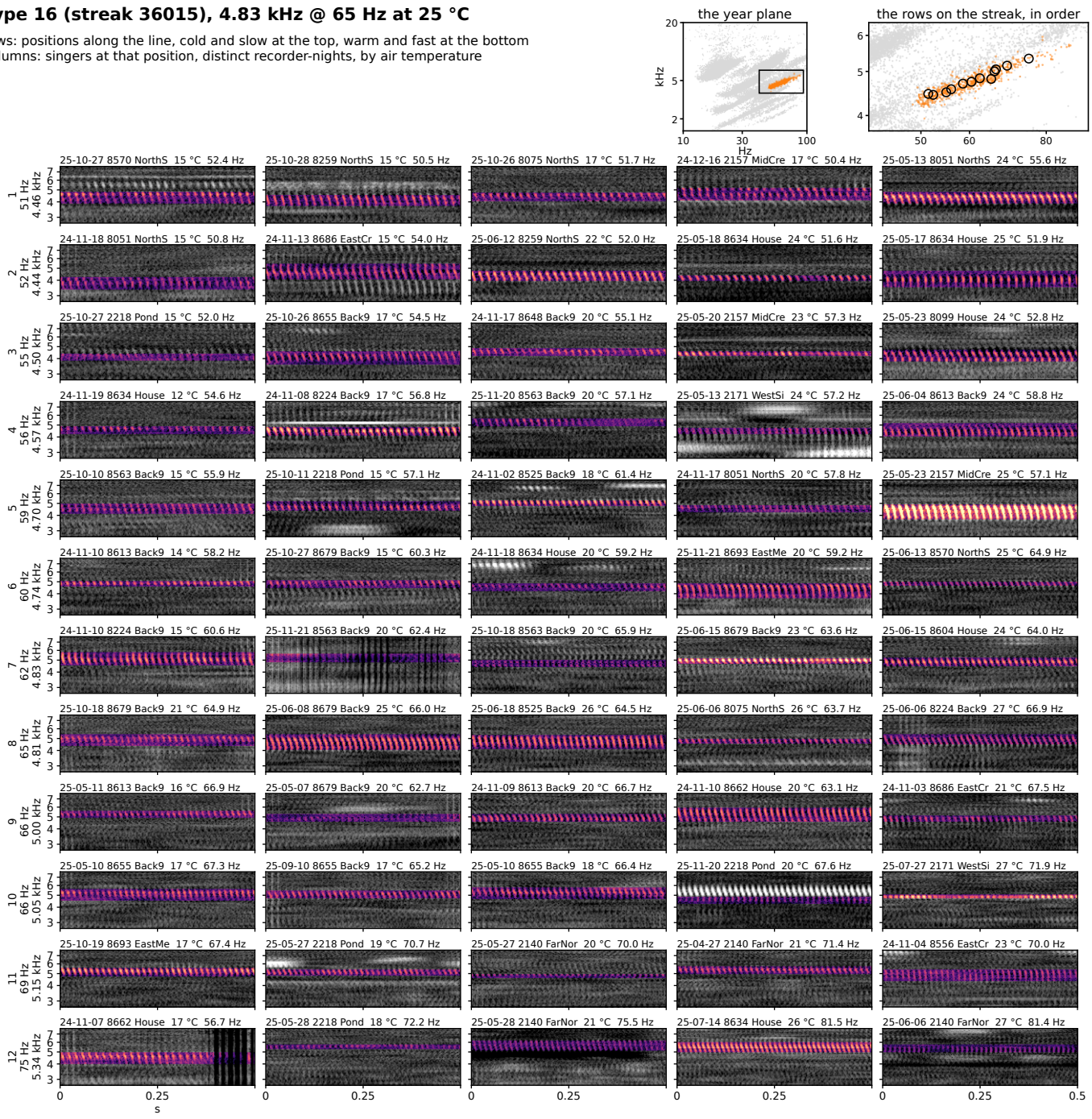

Figure S7-16 (rate-filtered). Type 16 (streak 36015): 4.83 kHz @ 65 Hz at 25 °C, 114 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

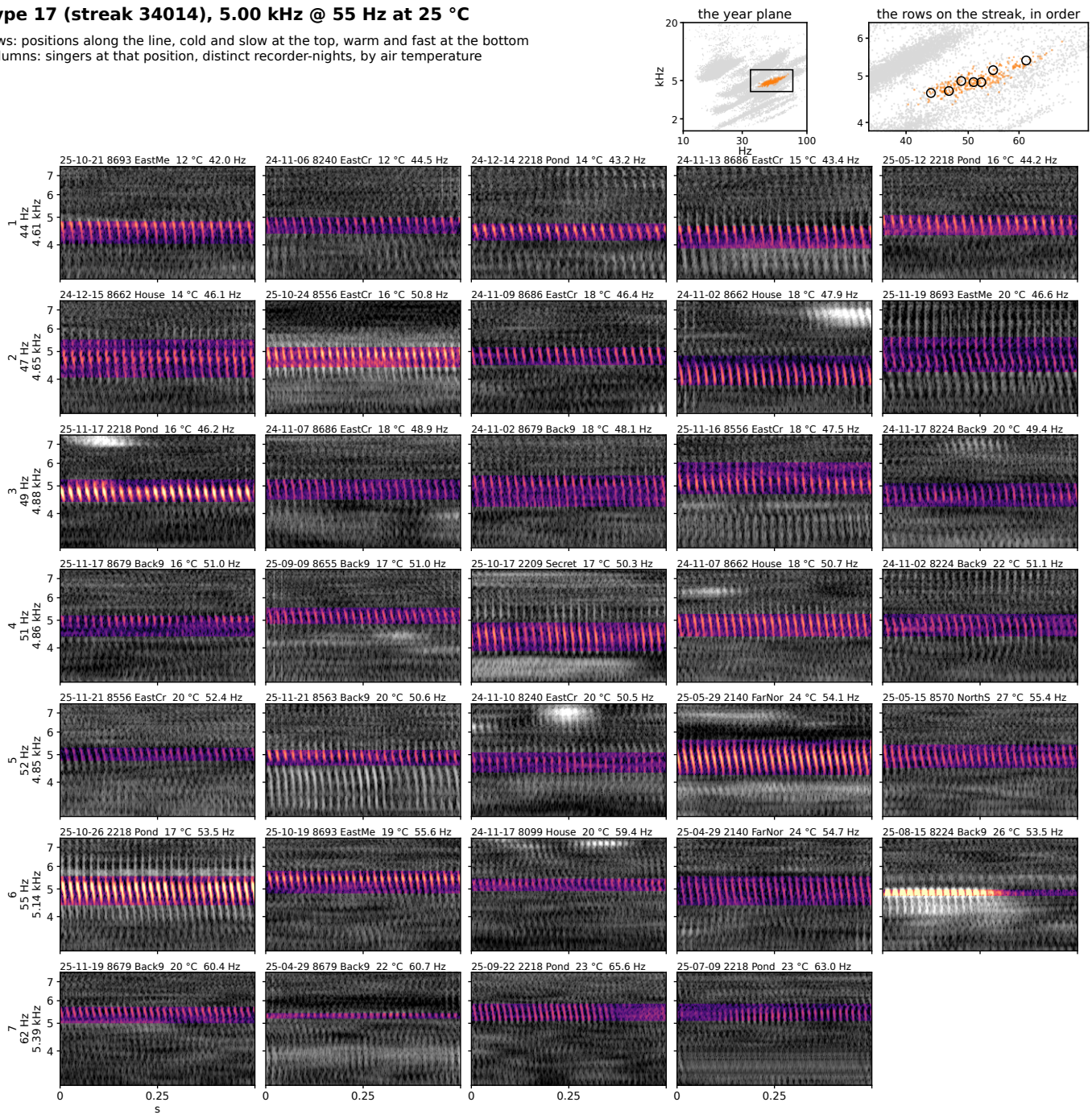

Figure S7-17 (rate-filtered). Type 17 (streak 34014): 5.00 kHz @ 55 Hz at 25 °C, 62 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

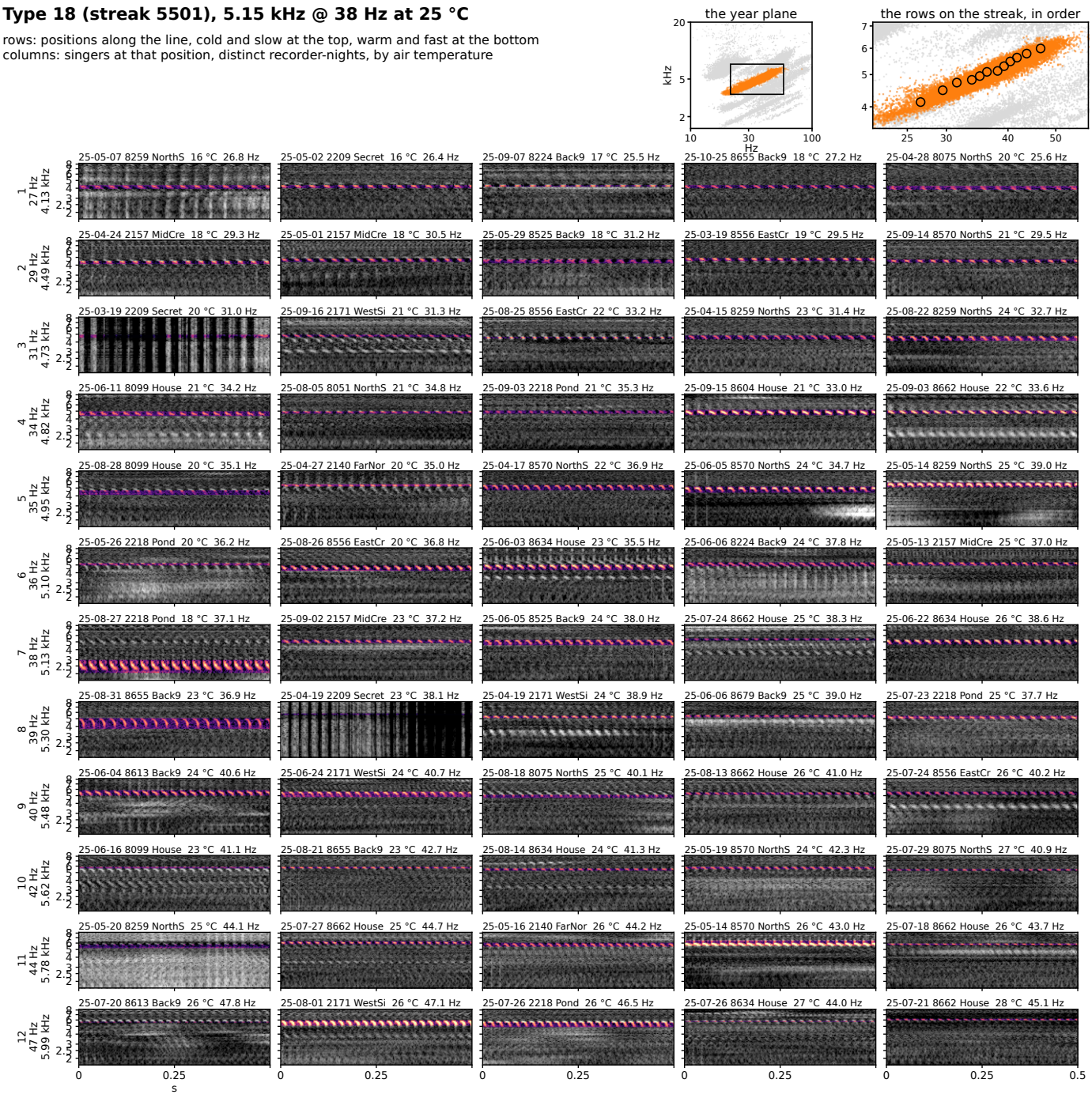

Figure S7-18 (rate-filtered). Type 18 (streak 5501): 5.15 kHz @ 38 Hz at 25 °C, 4213 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

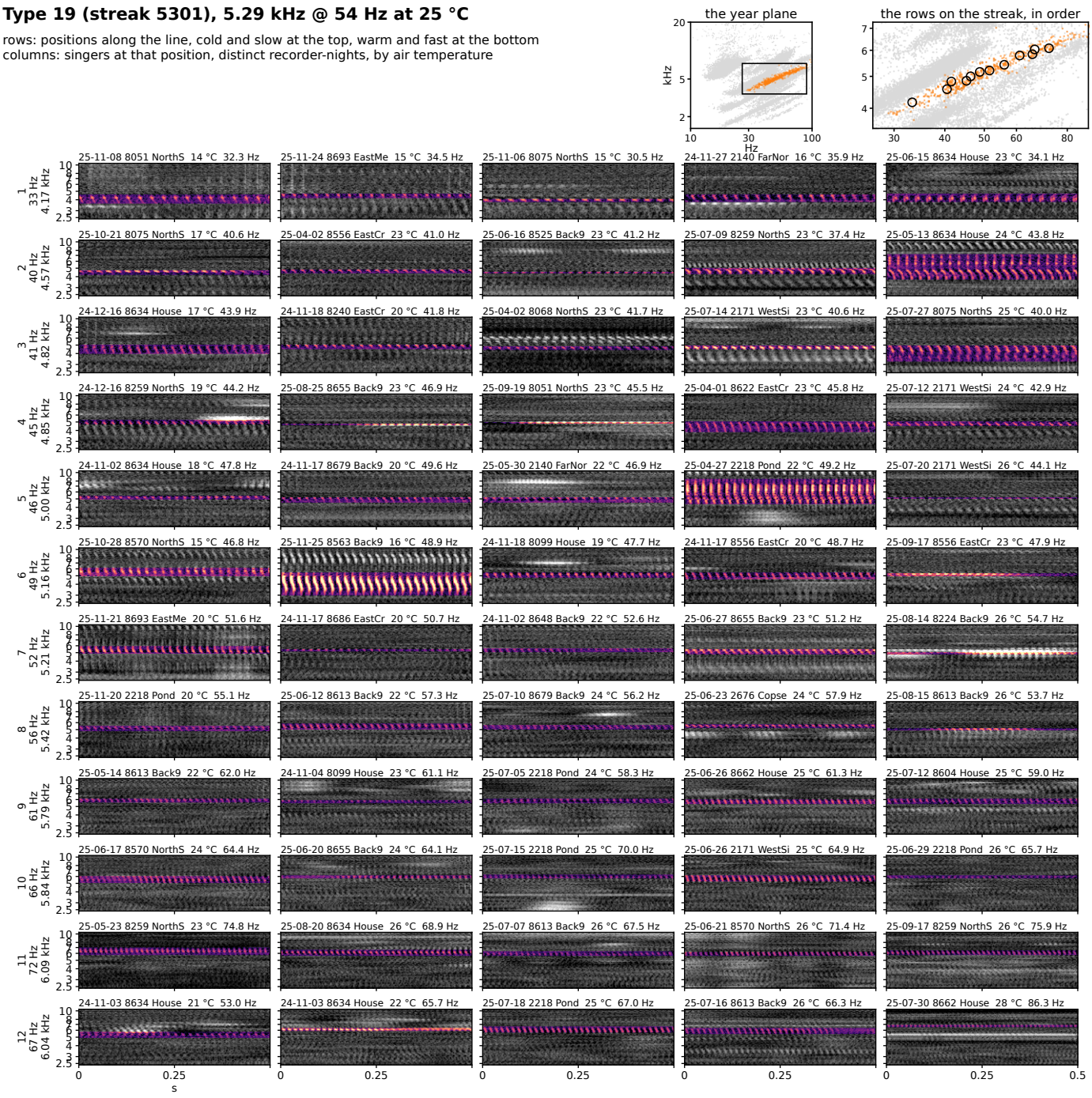

Figure S7-19 (rate-filtered). Type 19 (streak 5301): 5.29 kHz @ 54 Hz at 25 °C, 117 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

Type 20 (streak 3301), 5.34 kHz @ 69 Hz at 25 °C

rows: positions along the line, cold and slow at the top, warm and fast at the bottom
columns: singers at that position, distinct recorder-nights, by air temperature

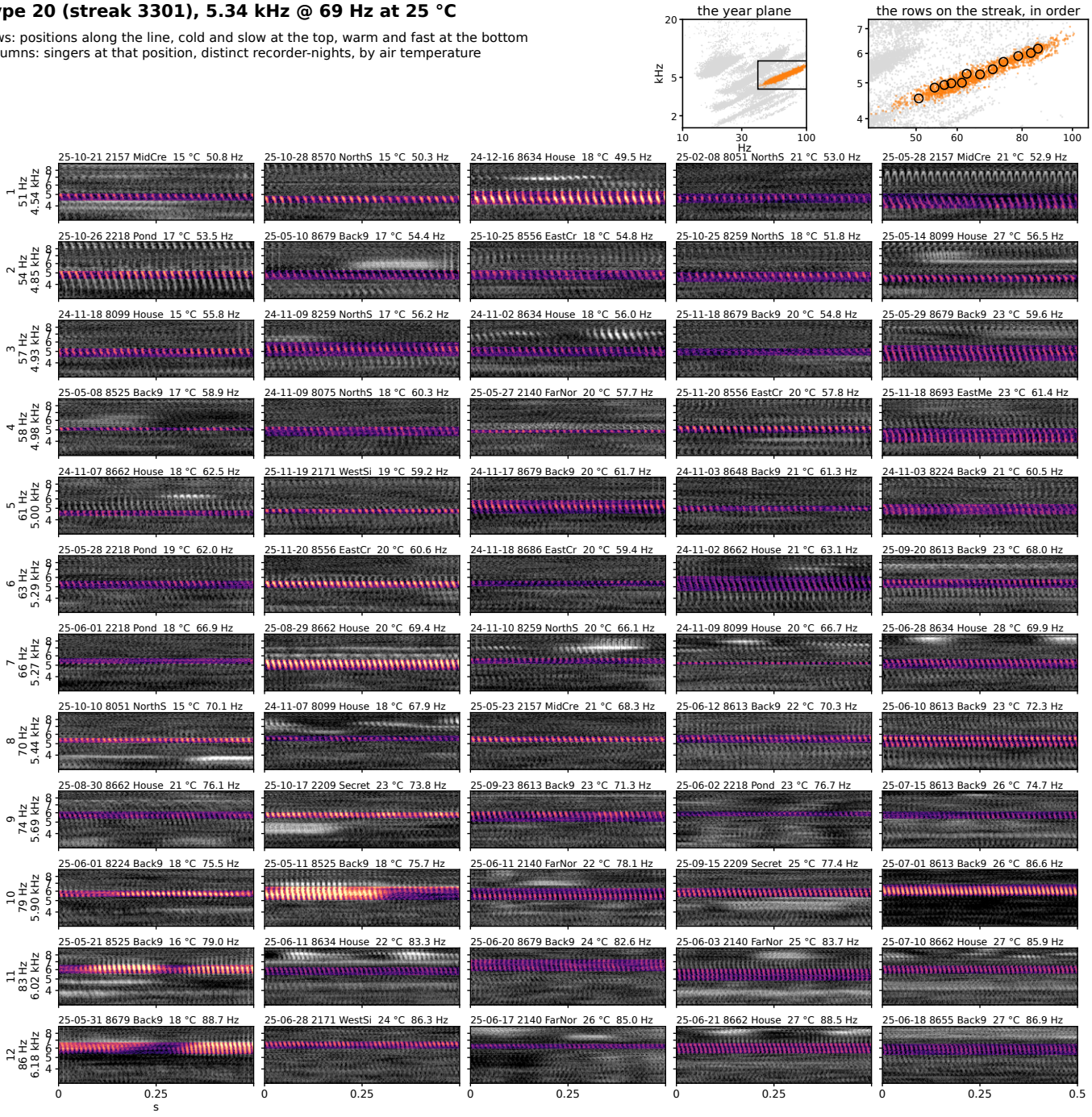

Figure S7-20 (rate-filtered). Type 20 (streak 3301): 5.34 kHz @ 69 Hz at 25 °C, 374 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

Type 21 (streak 6301), 5.66 kHz @ 20 Hz at 25 °C

rows: positions along the line, cold and slow at the top, warm and fast at the bottom
columns: singers at that position, distinct recorder-nights, by air temperature

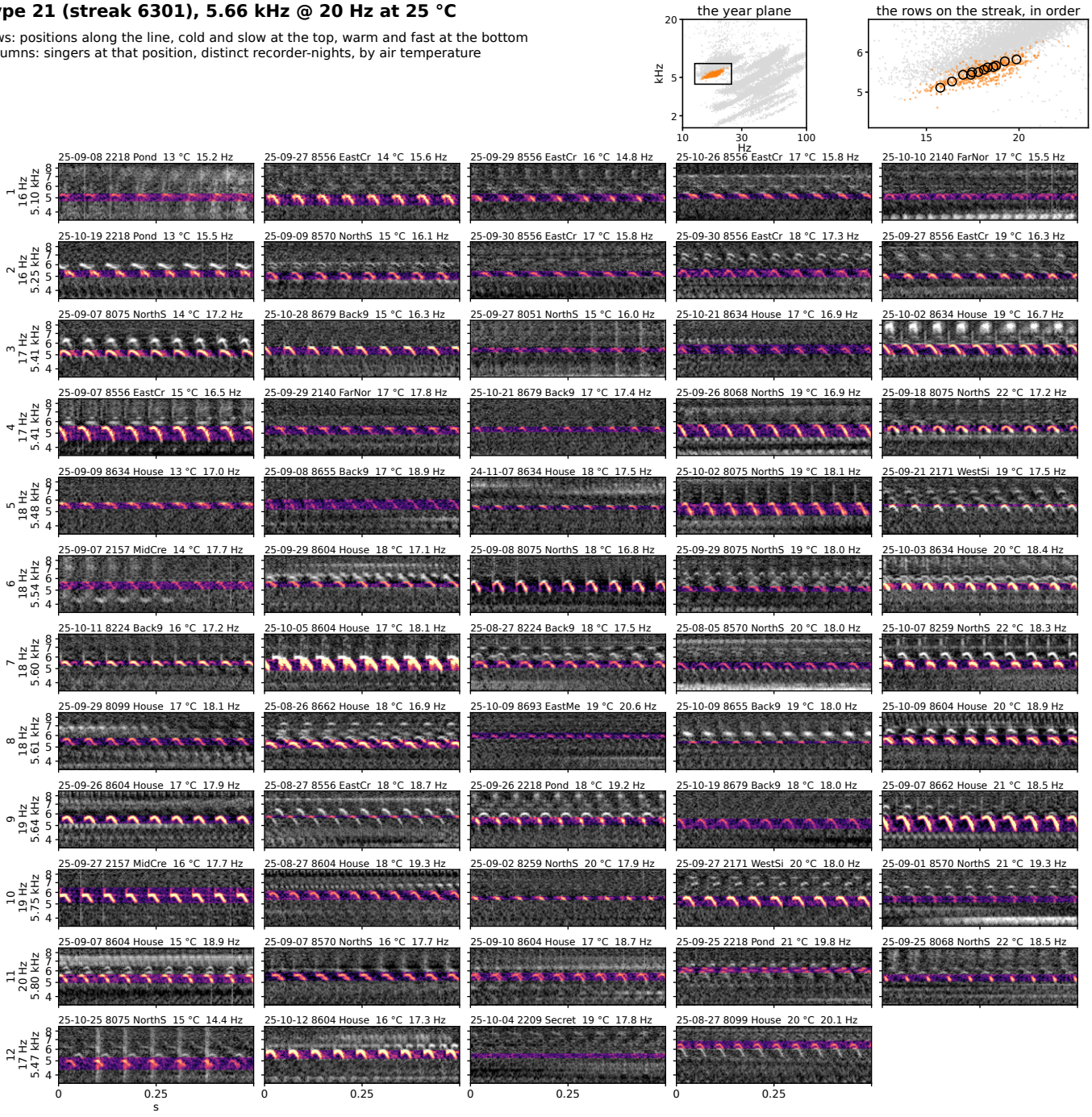

Figure S7-21 (rate-filtered). Type 21 (streak 6301): 5.66 kHz @ 20 Hz at 25 °C, 117 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

Type 22 (streak 590116), 5.86 kHz @ 28 Hz at 25 °C

rows: positions along the line, cold and slow at the top, warm and fast at the bottom
columns: singers at that position, distinct recorder-nights, by air temperature

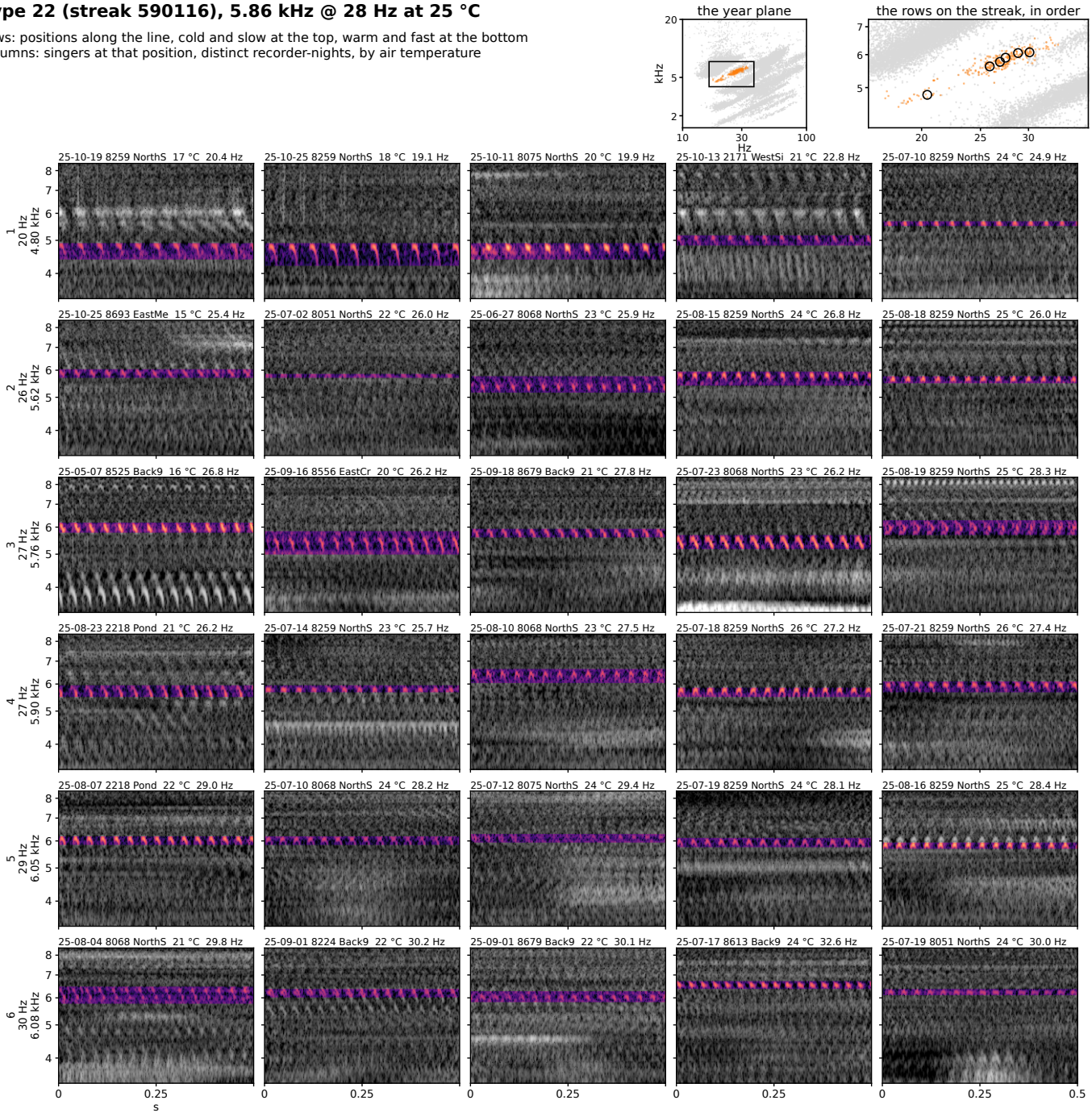

Figure S7-22 (rate-filtered). Type 22 (streak 590116): 5.86 kHz @ 28 Hz at 25 °C, 54 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

Type 23 (streak 5401), 6.00 kHz @ 43 Hz at 25 °C

rows: positions along the line, cold and slow at the top, warm and fast at the bottom
columns: singers at that position, distinct recorder-nights, by air temperature

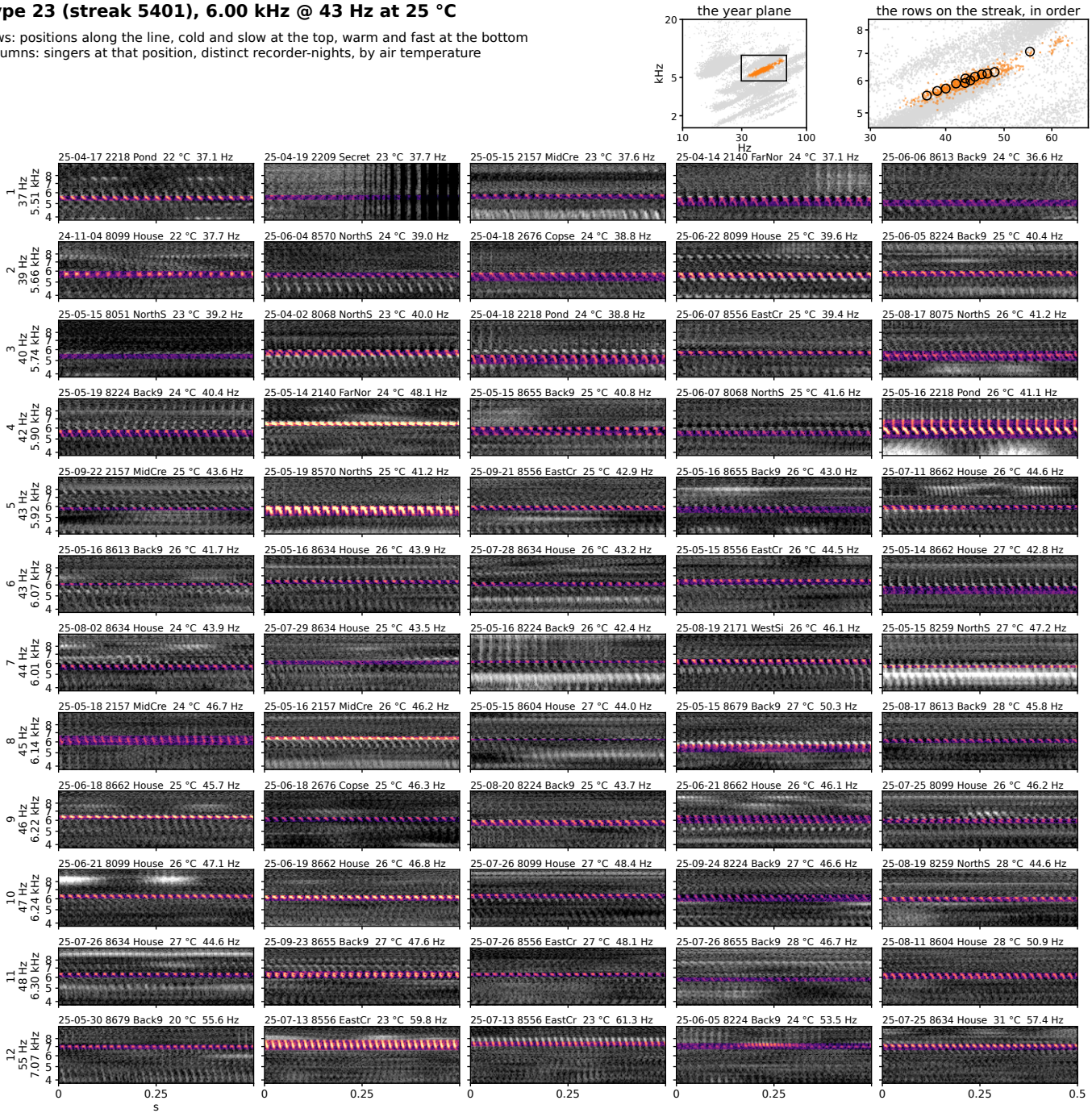

Figure S7-23 (rate-filtered). Type 23 (streak 5401): 6.00 kHz @ 43 Hz at 25 °C, 143 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

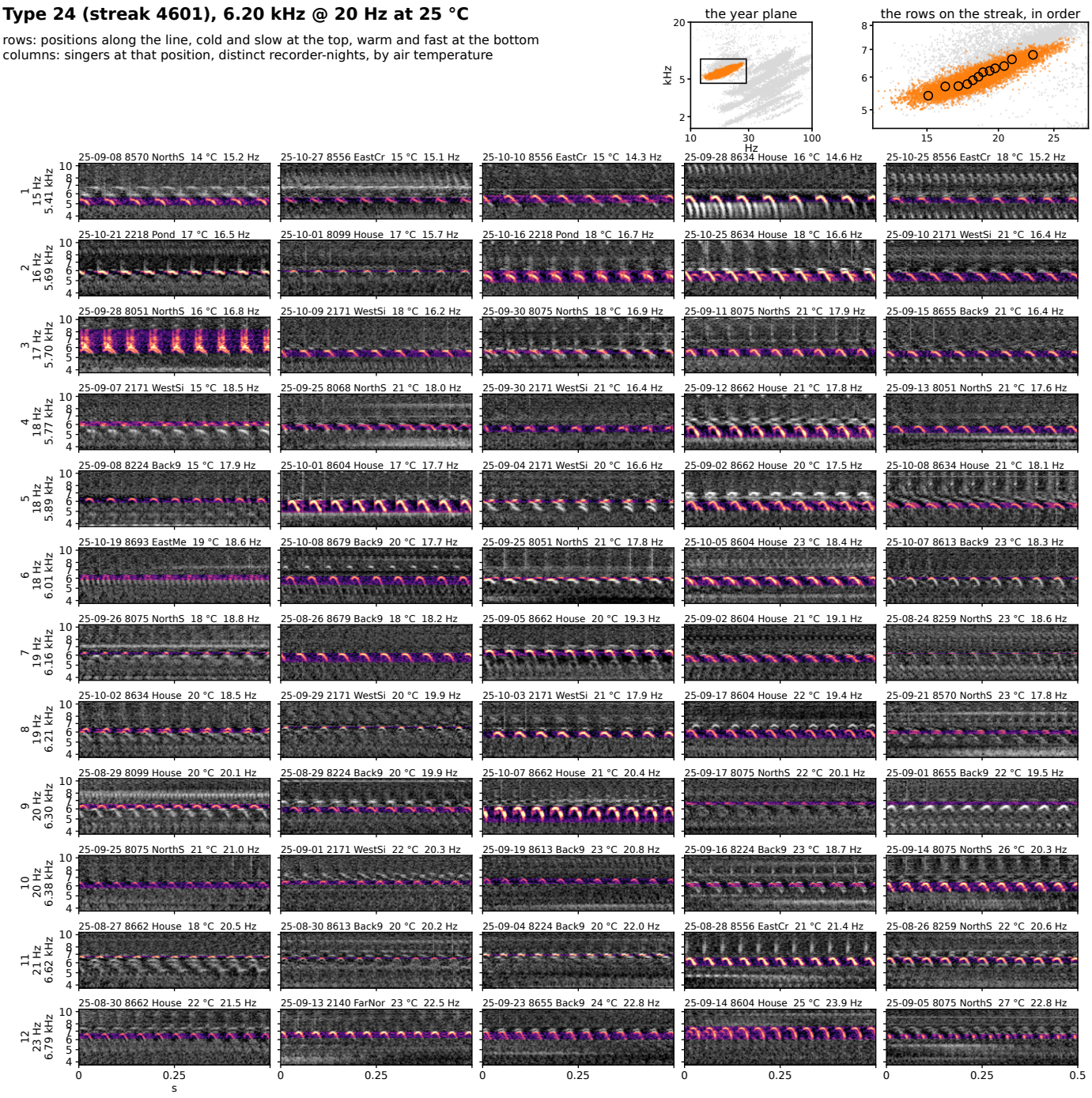

Figure S7-24 (rate-filtered). Type 24 (streak 4601): 6.20 kHz @ 20 Hz at 25 °C, 1908 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

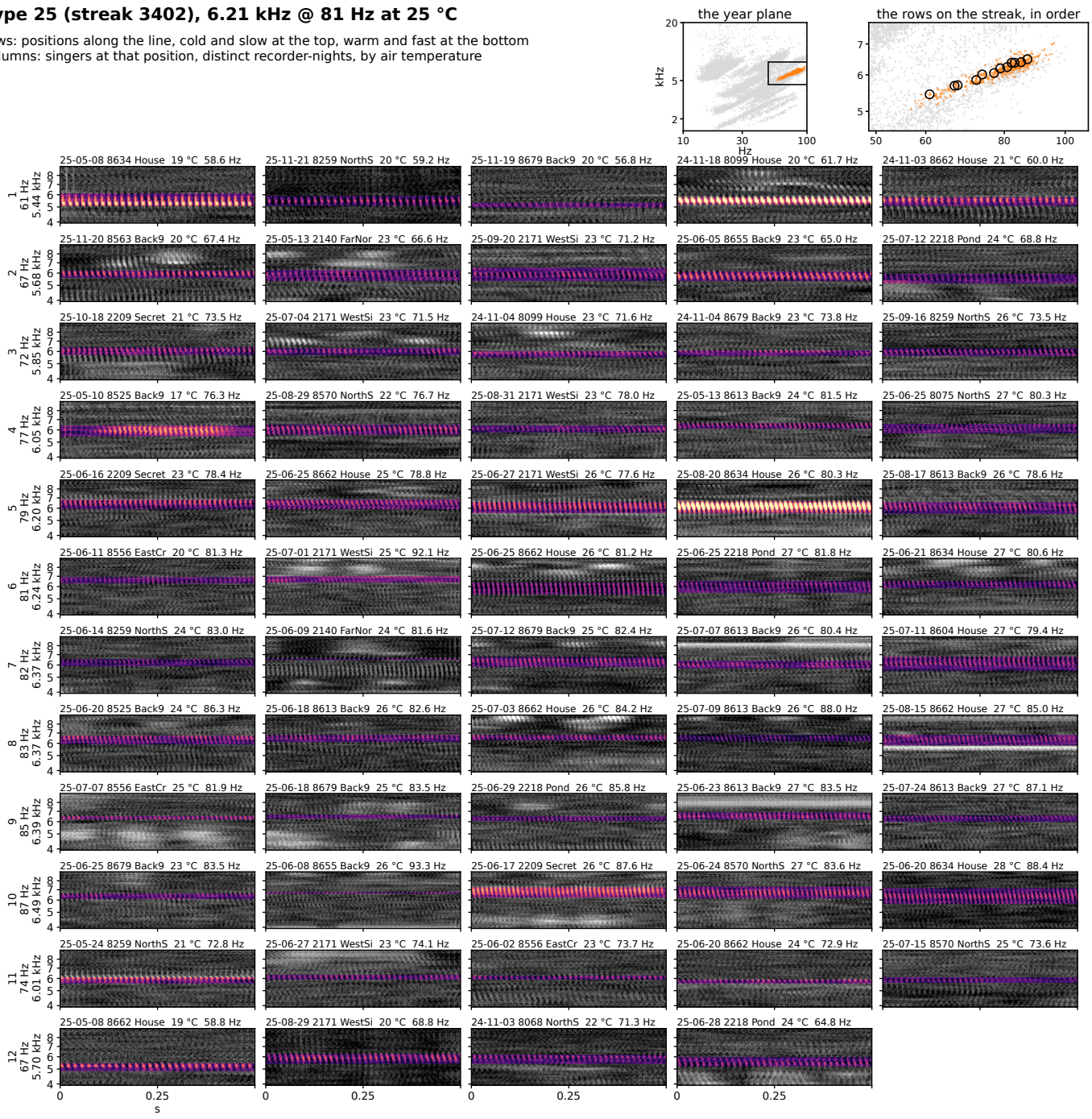

Figure S7-25 (rate-filtered). Type 25 (streak 3402): 6.21 kHz @ 81 Hz at 25 °C, 105 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

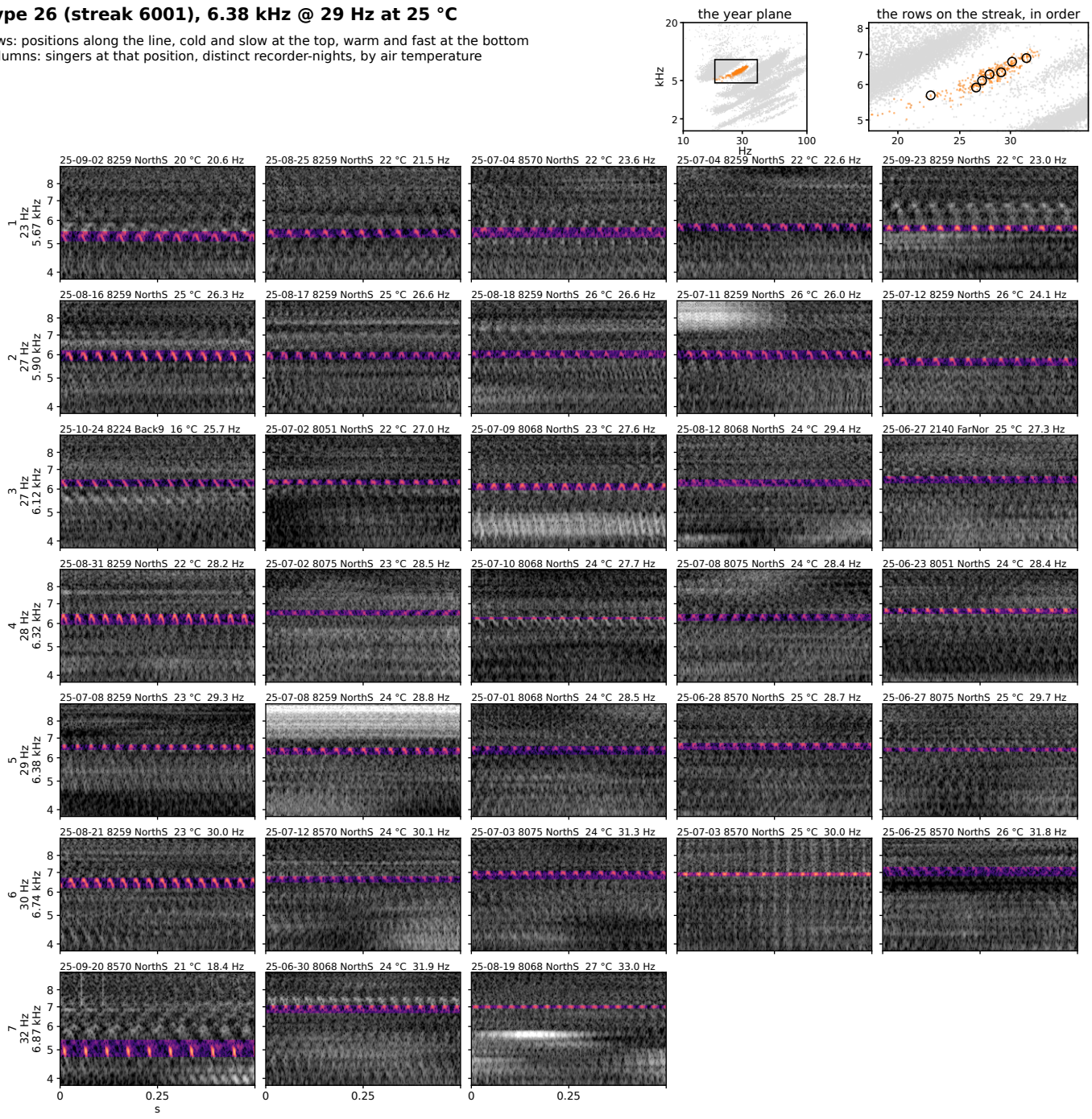

Figure S7-26 (rate-filtered). Type 26 (streak 6001): 6.38 kHz @ 29 Hz at 25 °C, 61 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

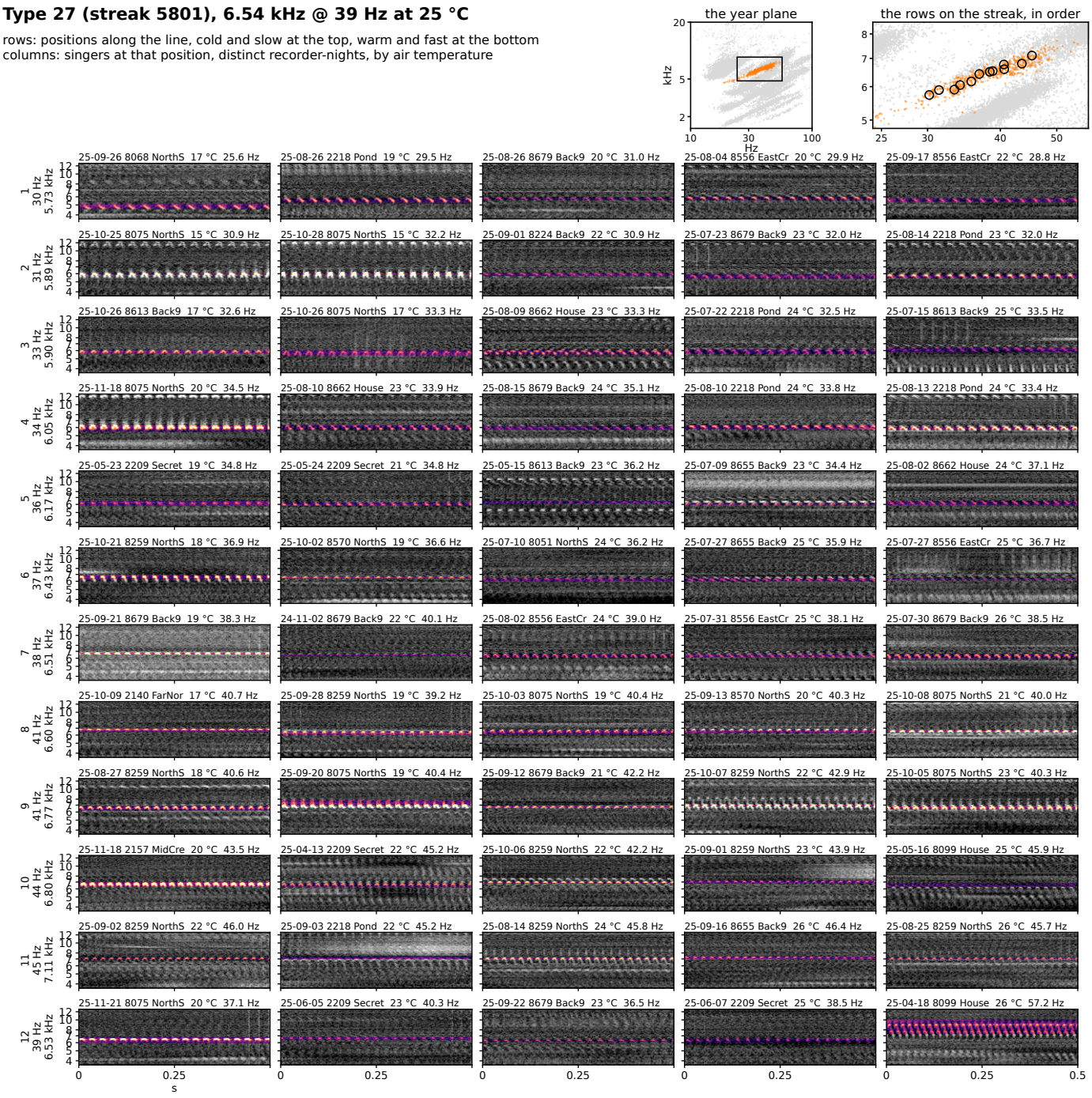

Figure S7-27 (rate-filtered). Type 27 (streak 5801): 6.54 kHz @ 39 Hz at 25 °C, 124 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

Type 28 (streak 590117), 6.95 kHz @ 38 Hz at 25 °C

rows: positions along the line, cold and slow at the top, warm and fast at the bottom
columns: singers at that position, distinct recorder-nights, by air temperature

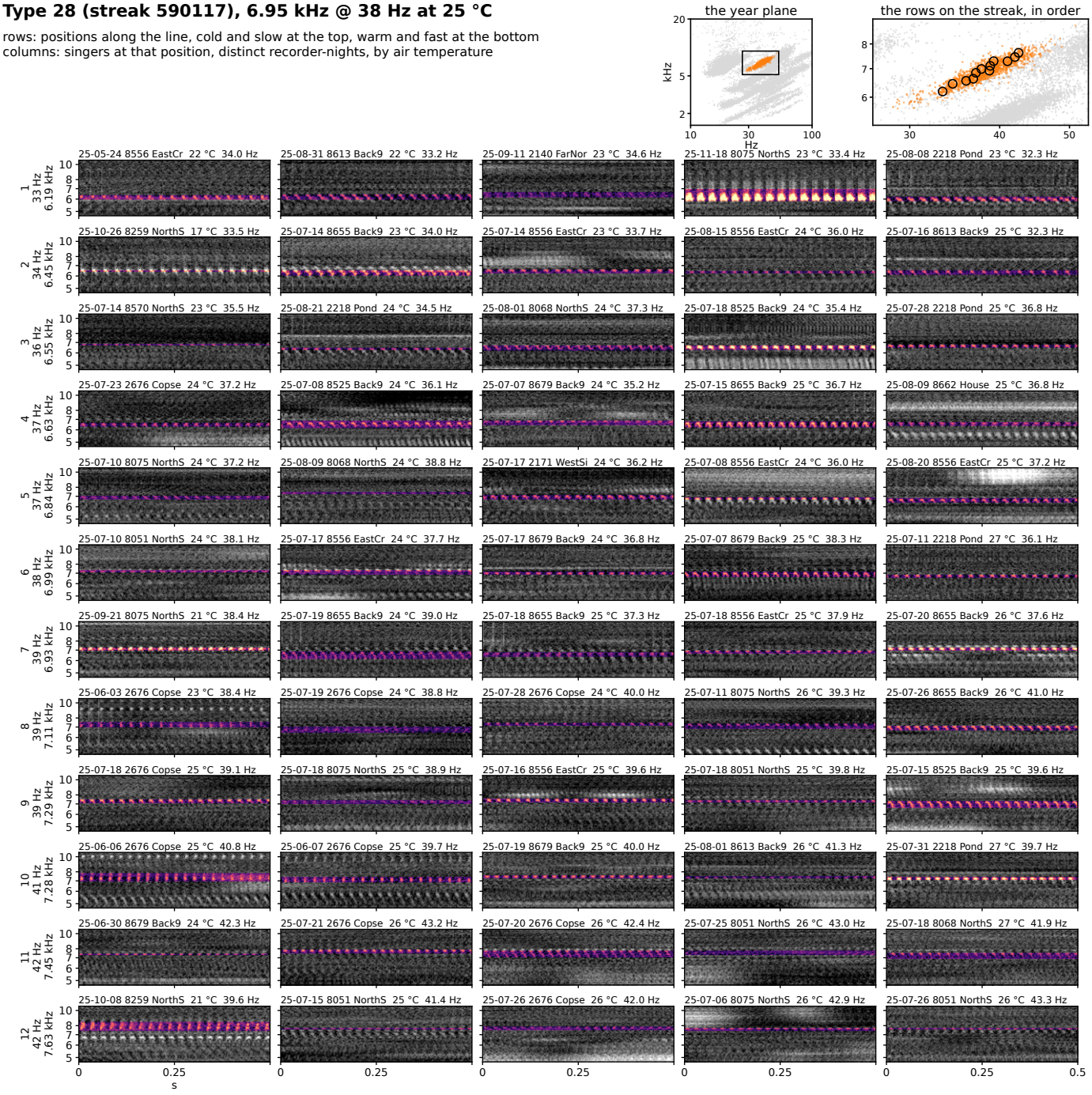

Figure S7-28 (rate-filtered). Type 28 (streak 590117): 6.95 kHz @ 38 Hz at 25 °C, 273 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

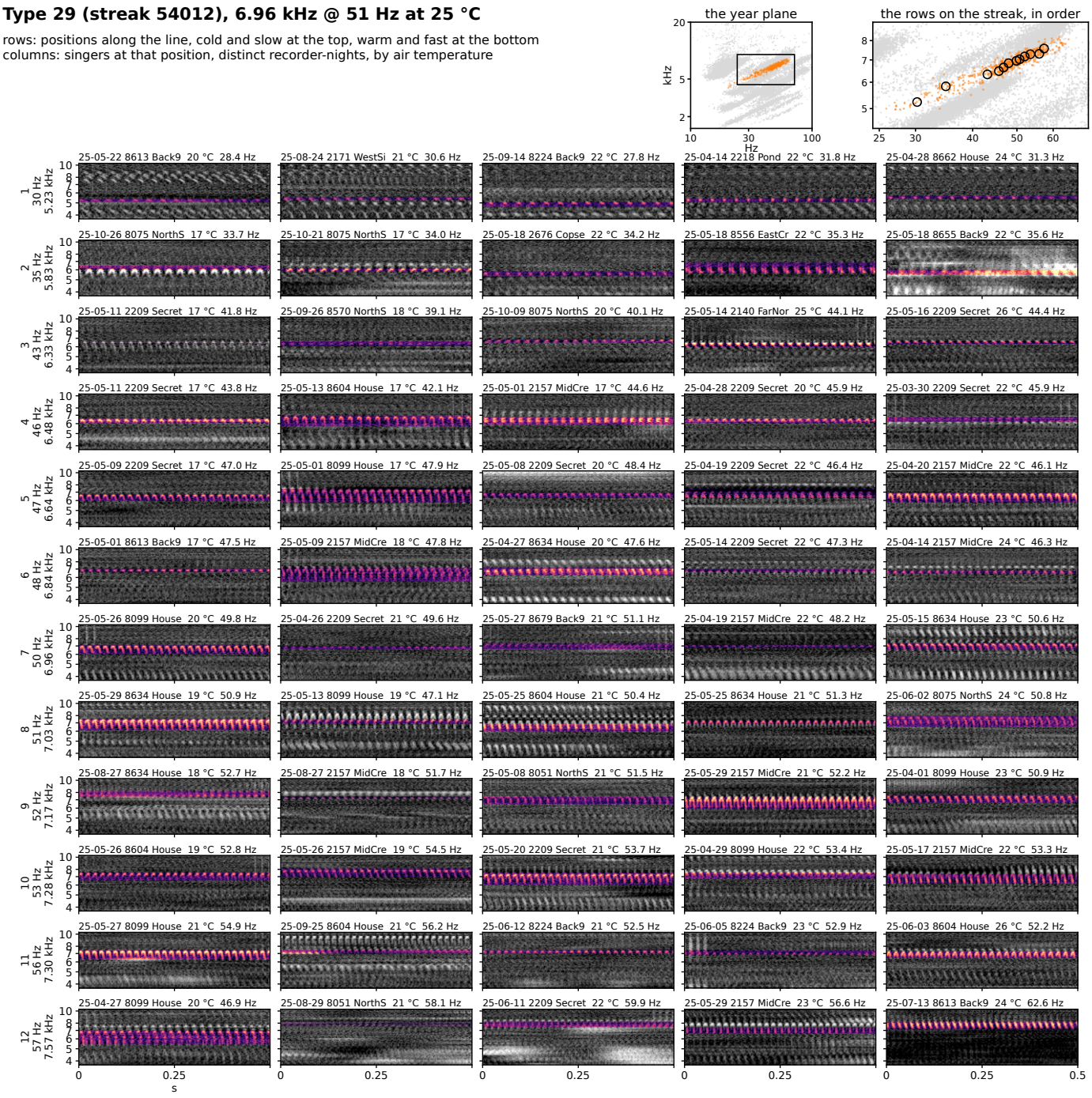

Figure S7-29 (rate-filtered). Type 29 (streak 54012): 6.96 kHz @ 51 Hz at 25 °C, 128 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

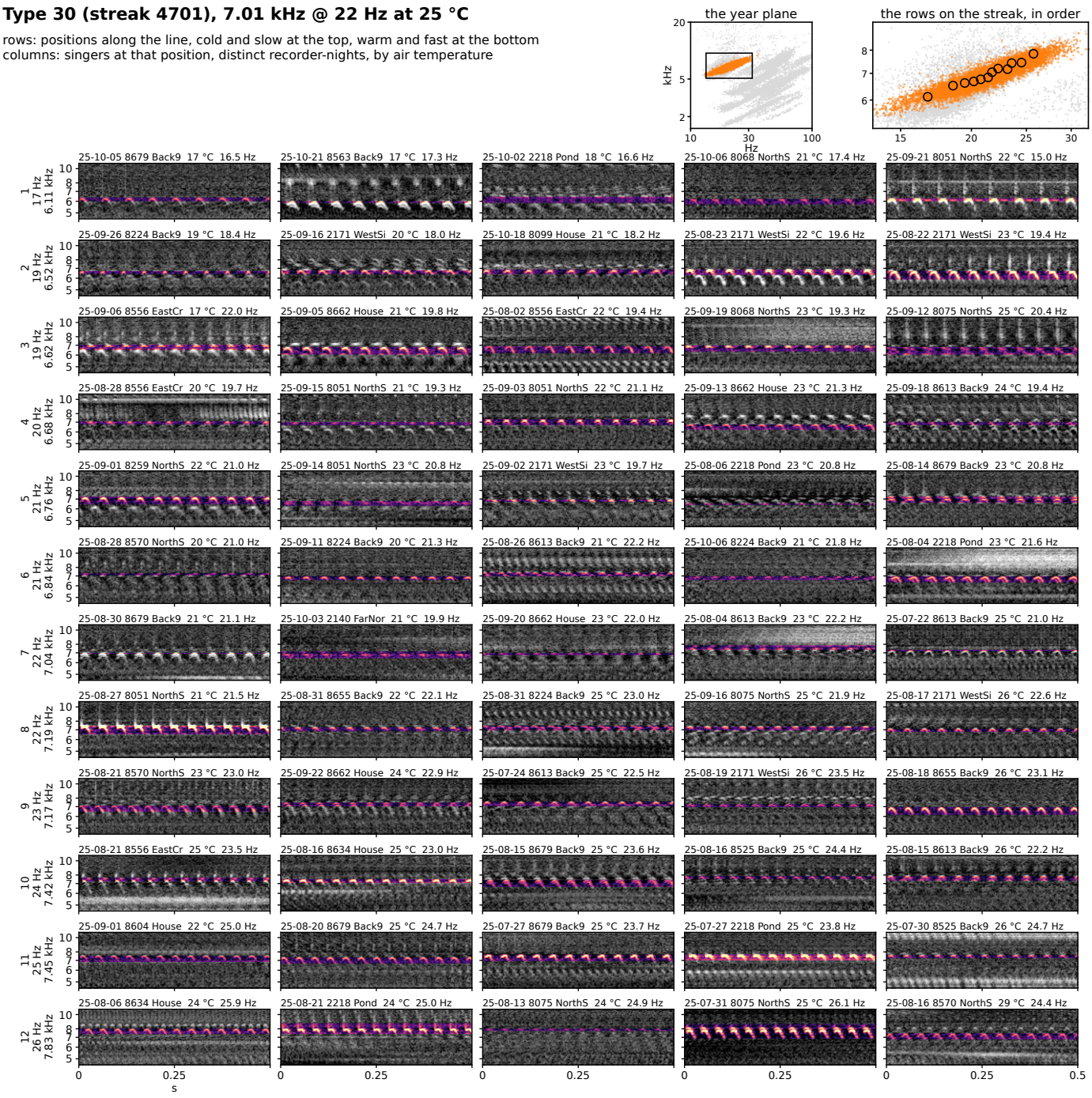

Figure S7-30 (rate-filtered). Type 30 (streak 4701): 7.01 kHz @ 22 Hz at 25 °C, 2080 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

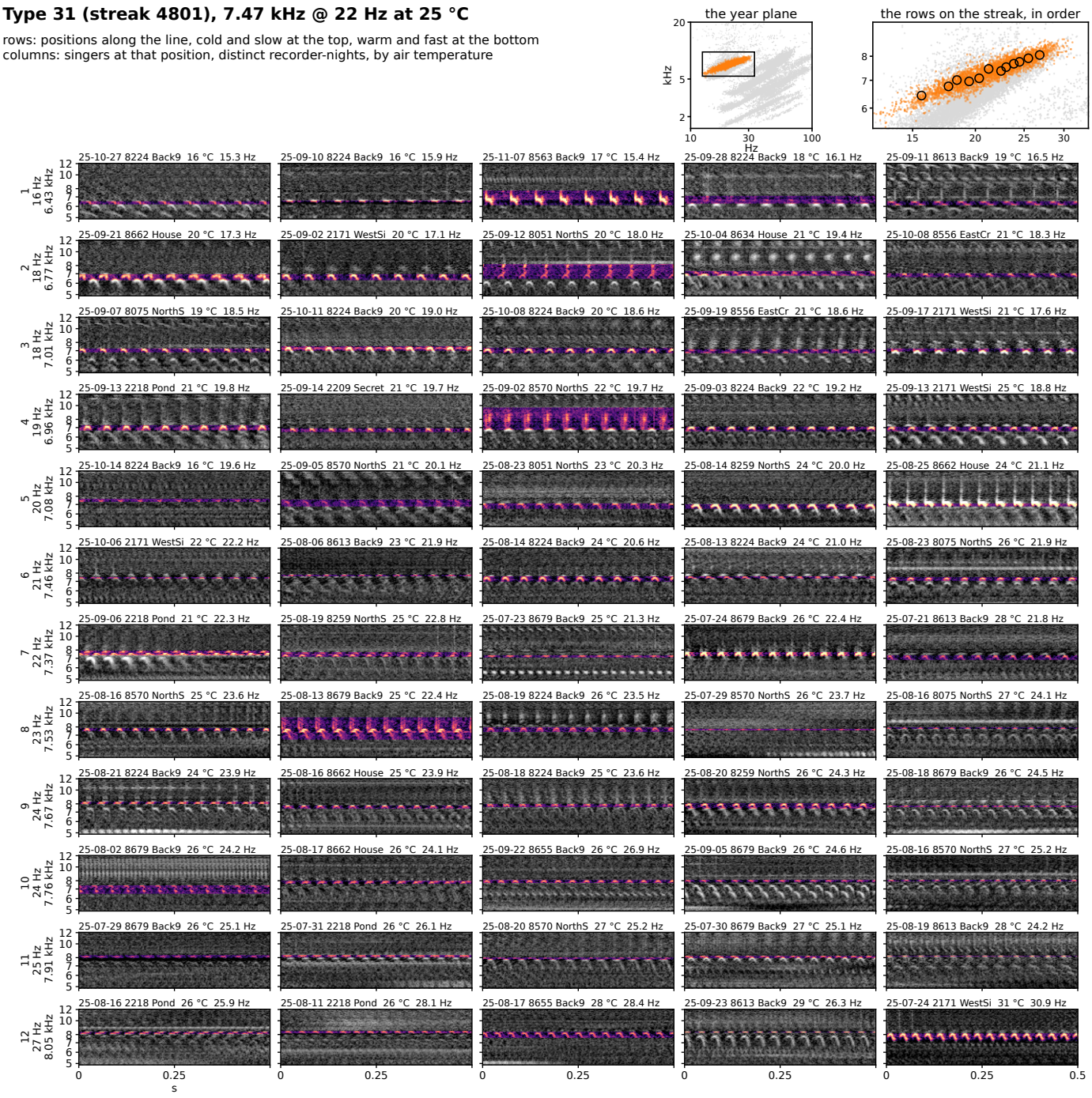

Figure S7-31 (rate-filtered). Type 31 (streak 4801): 7.47 kHz @ 22 Hz at 25 °C, 623 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

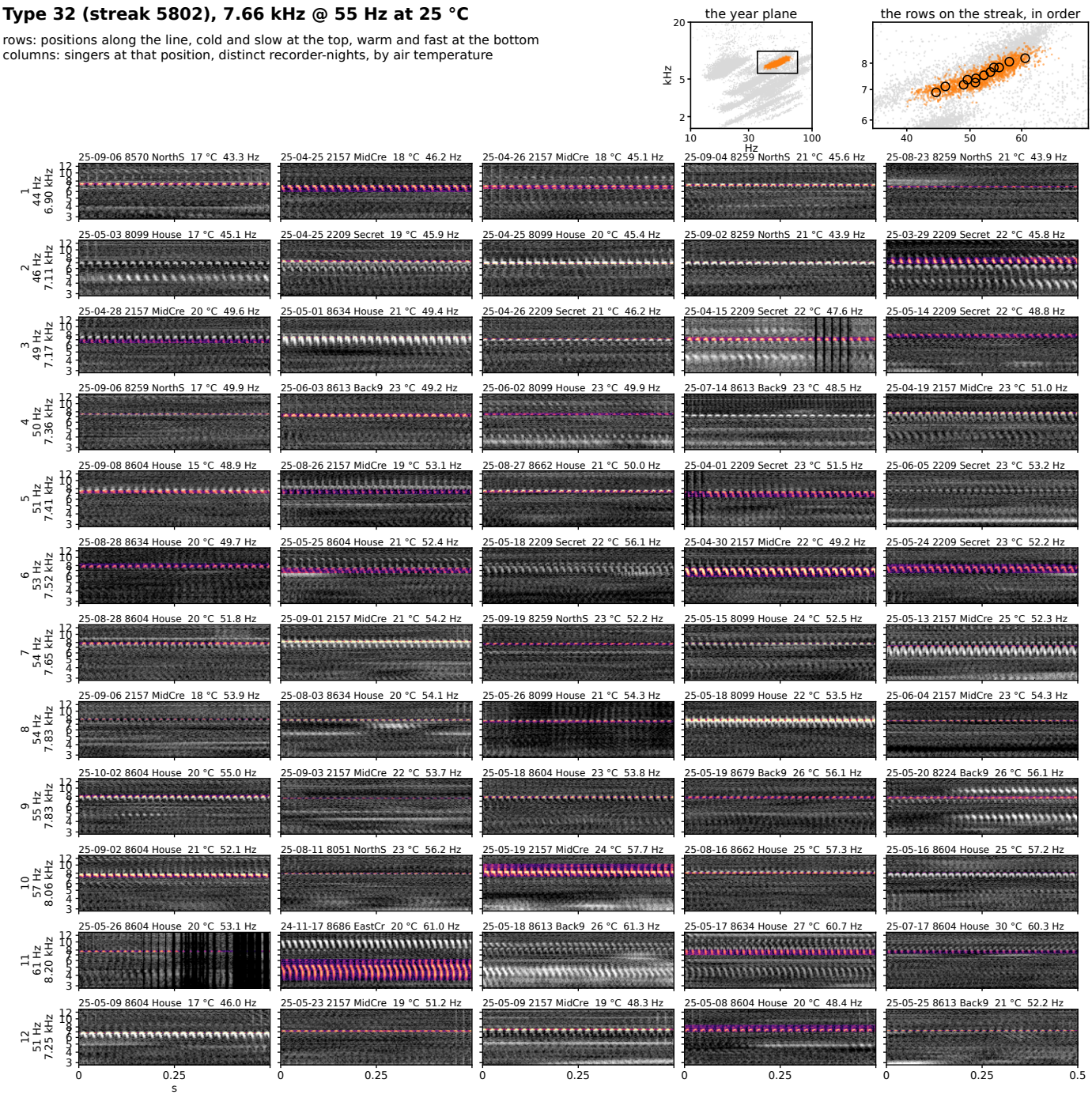

Figure S7-32 (rate-filtered). Type 32 (streak 5802): 7.66 kHz @ 55 Hz at 25 °C, 375 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

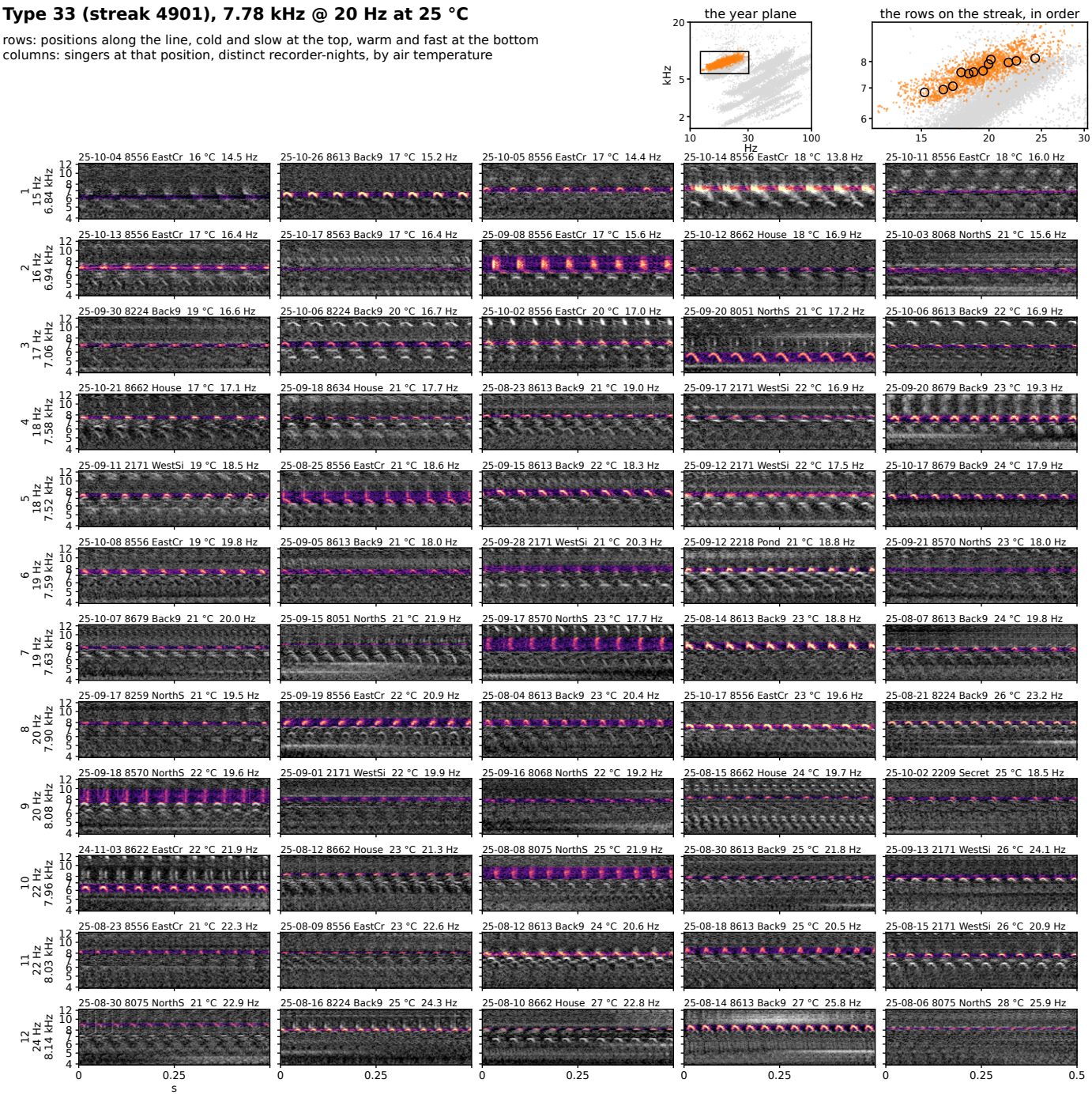

Figure S7-33 (rate-filtered). Type 33 (streak 4901): 7.78 kHz @ 20 Hz at 25 °C, 391 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

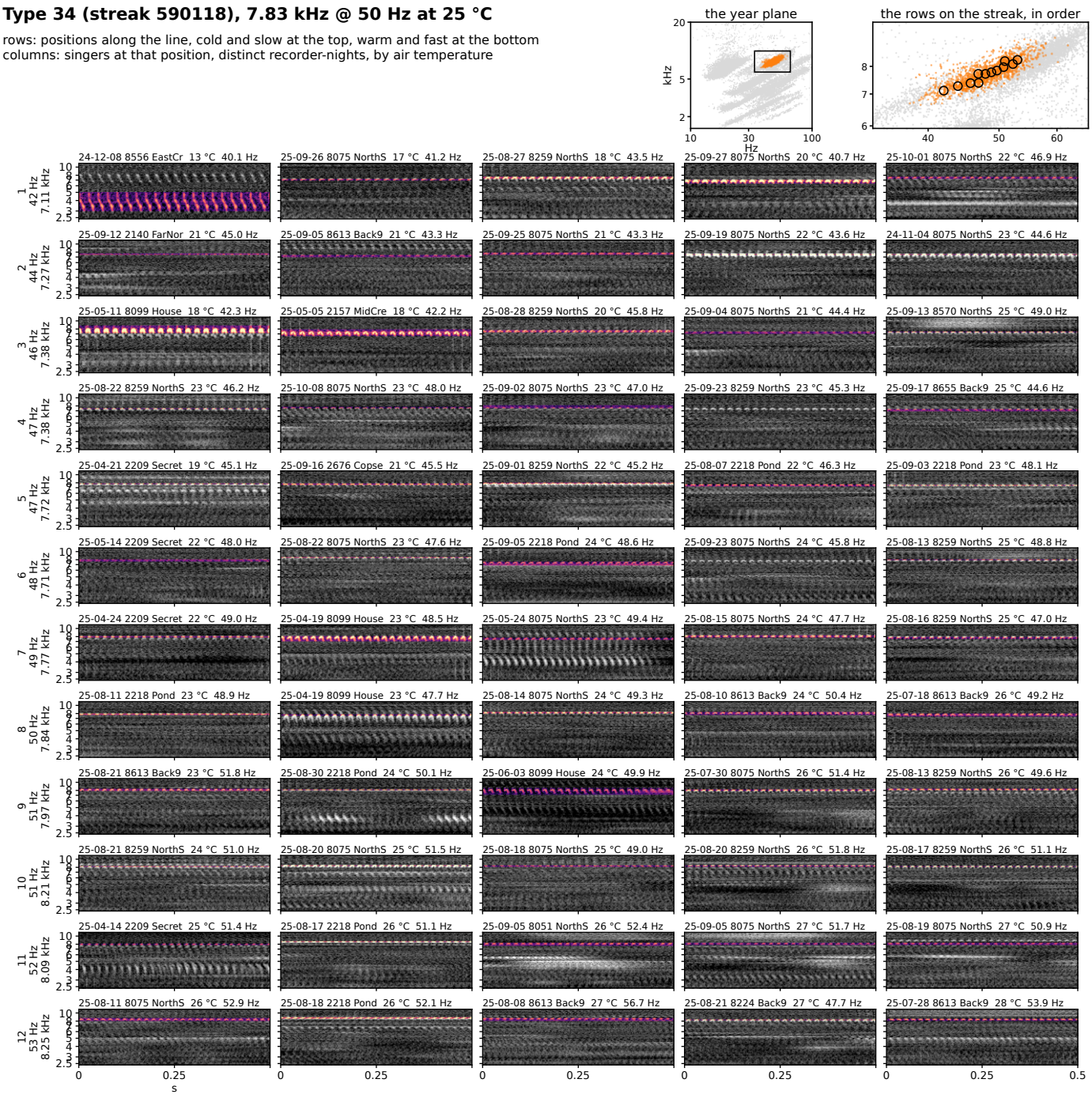

Figure S7-34 (rate-filtered). Type 34 (streak 590118): 7.83 kHz @ 50 Hz at 25 °C, 300 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

Type 35 (streak 5901), 8.55 kHz @ 58 Hz at 25 °C

rows: positions along the line, cold and slow at the top, warm and fast at the bottom
columns: singers at that position, distinct recorder-nights, by air temperature

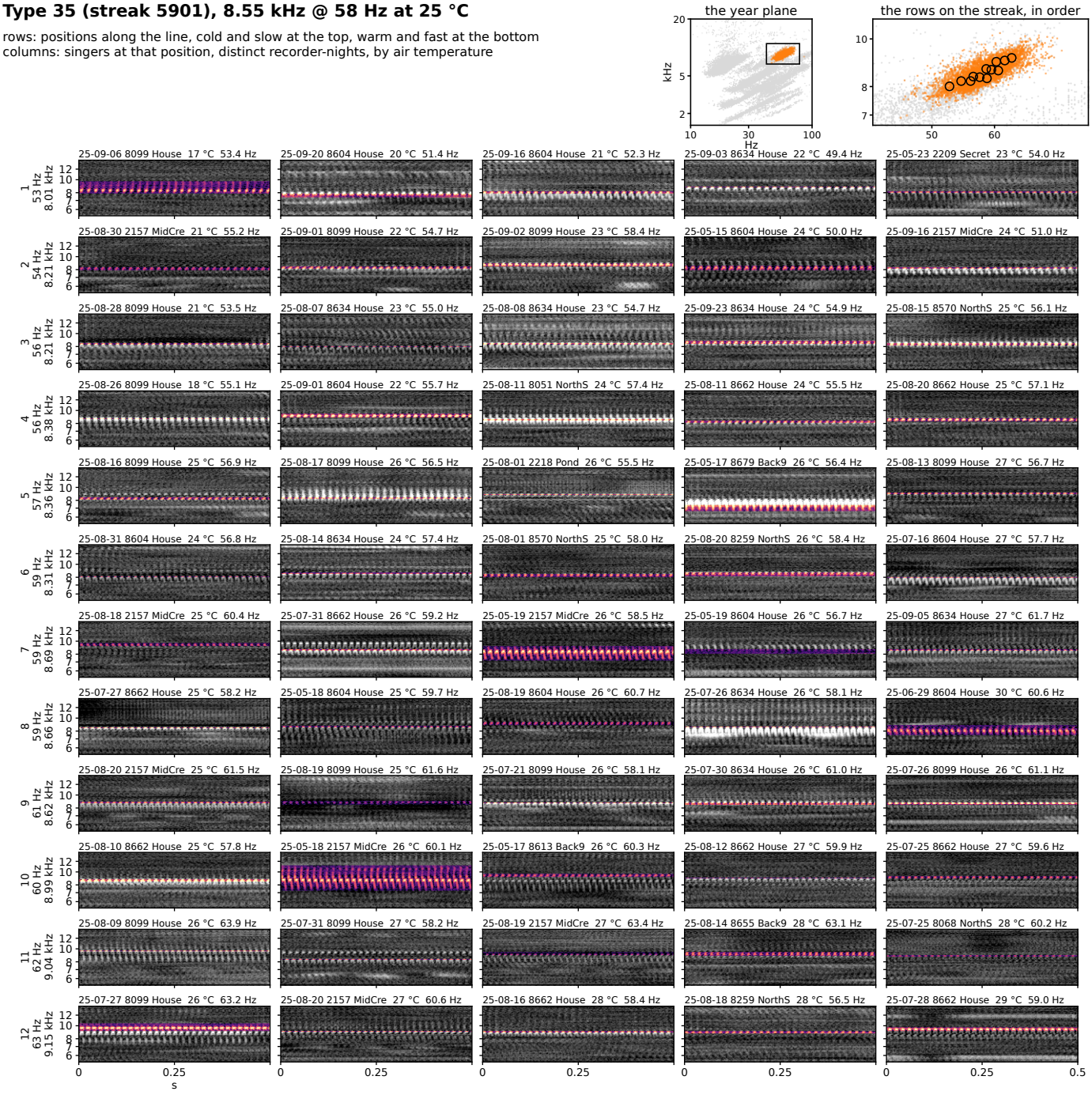

Figure S7-35 (rate-filtered). Type 35 (streak 5901): 8.55 kHz @ 58 Hz at 25 °C, 826 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

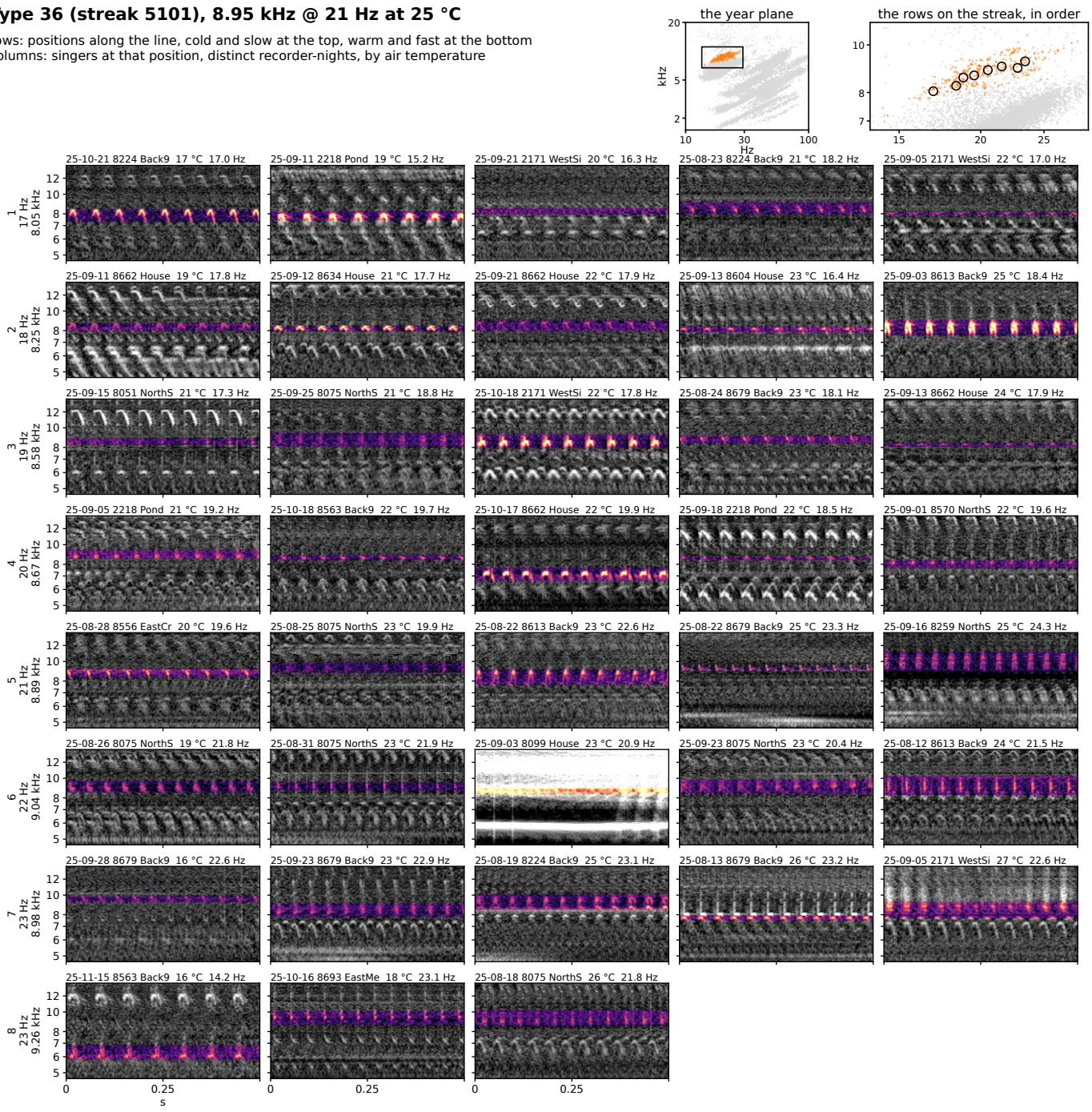

Figure S7-36 (rate-filtered). Type 36 (streak 5101): 8.95 kHz @ 21 Hz at 25 °C, 69 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.

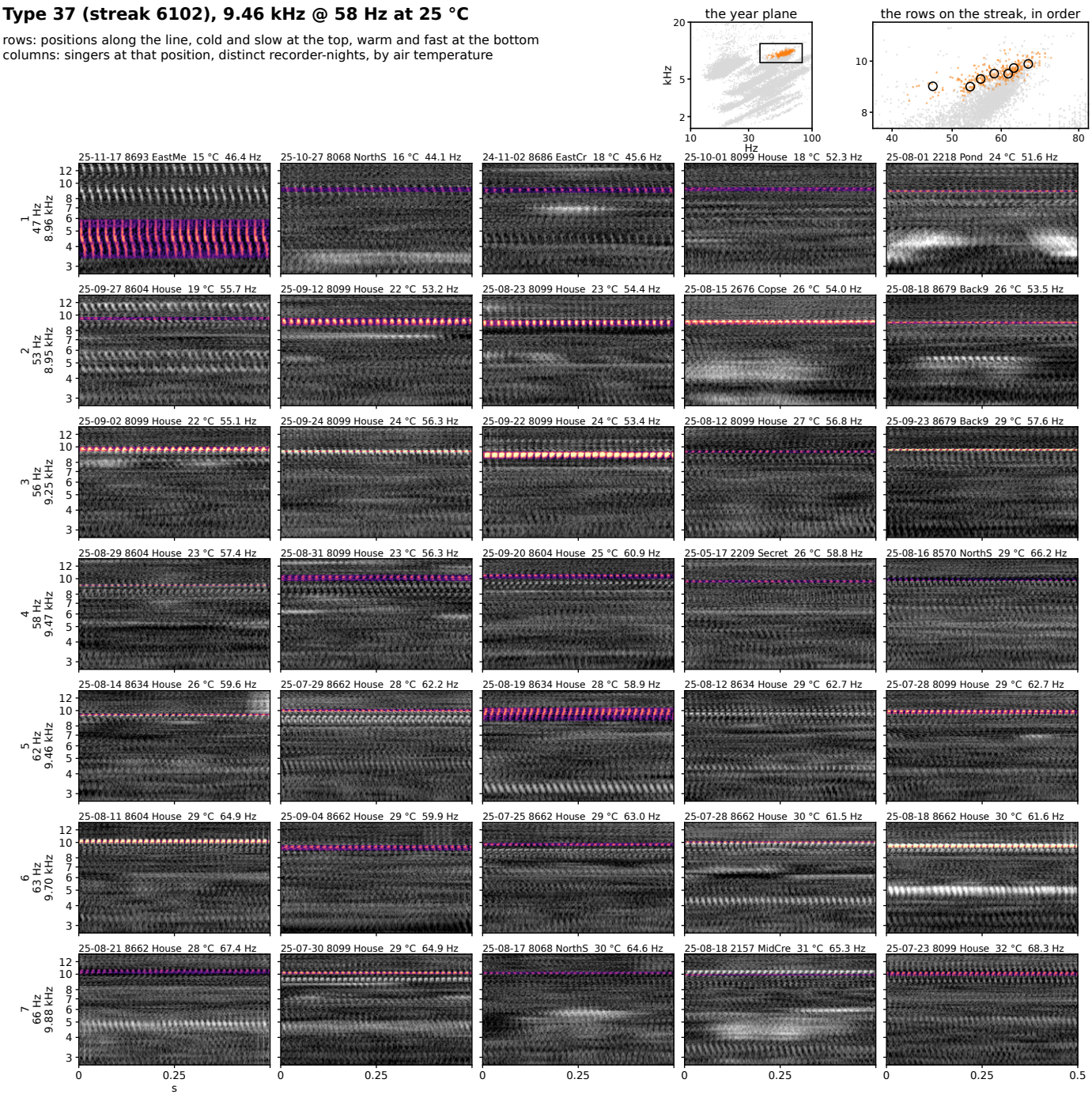

Figure S7-37 (rate-filtered). Type 37 (streak 6102): 9.46 kHz @ 58 Hz at 25 °C, 64 singer-nights. RATE-FILTERED VIEW of the grid: each cell is the period-synchronous average of the excerpt over 5 periods at the singer's own rate, so the singer's syllables stand out and neighbors at other rates smear. Rows are positions along the line from the cold, slow end at the top to the warm, fast end at the bottom; columns are singers at that position from distinct recorder-nights, ordered by air temperature. Each cell is 0.5 s of the loudest 30 s of the singer's deepest window on one shared log-frequency axis, the detected band in color (dB above the row's quiet level, 0 to 30) and the rest in grayscale. The cell head gives the night, recorder, site, air temperature and rate (c: chirp rate). Top right: the type on the plane, and the row positions on the streak, in order.